



TECHNICAL MEMORANDUM:
STANDARD FORMAT FOR THE ELECTRONIC
TRANSFER OF LUMINAIRE OPTICAL DATA
AN AMERICAN NATIONAL STANDARD



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Publication of this Technical Memorandum
has been approved by IES.
Suggestions for revisions
should be directed to IES.

Prepared by the the IES Computer Committee



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Preface

This preface is not part of ANSI/IES TM-33-23. It is included for informational purposes only.

Updates in the 2023 version:

- Angular Color

An `AngularColor` complex element (**Section 4.5.18**) has been added.

- Horizontal Symmetry

A `Symm` element has been added to these sections:

- 4.5.14.1 Luminous Intensity
- 4.5.15.3 Radiant Intensity
- 4.5.16.3 Photon Intensity
- 4.5.17.2 Angular Spectral
- 4.5.18.2 Angular Color

- Multiplier

A `Multiplier` element has been added to these sections:

- 4.5.14.1 Luminous Intensity
- 4.5.15.3 Radiant Intensity
- 4.5.16.3 Photon Intensity
- 4.5.17.1 Emitter Spectral
- 4.5.17.2 Angular Spectral
- 4.5.18 Angular Color
- 4.5.19 Illuminance
- 4.5.20 Irradiance
- 4.5.21 Photon Flux Density
- 4.5.22 Spectral Irradiance

- Absolute

An `Absolute` element has been added to these sections:

- 4.5.15.1 Radiant Intensity
- 4.5.16.1 Photon Intensity
- 4.5.17.2 Angular Spectral
- 4.5.18.1 Angular Color
- 4.5.19.1 Illuminance
- 4.5.20.1 Irradiance
- 4.5.21.1 Photon Flux Density
- 4.5.22.1 Spectral Irradiance

- Normalized

A `Normalized` element has been added to **Section 4.5.17.1 Emitter Spectral**.

- Timestamp

A `Timestamp` element has been added to **Section 4.5.17.1 Emitter Spectral**.

- Quantum

A `Quantum` element has been added to **Section 4.5.17 Emitter Spectral**.

- JSON Document Format

An equivalent JSON specification to the XML document specification is shown in **Section 4.8 JSON Document Format**.

1.0 Introduction and Scope

1.1 Introduction

The architectural and roadway lighting communities have long relied on standardized data formats for the electronic transfer of far-field photometric data and related information. These data formats include IES LM-63-02, ANSI/IES LM-63-19, CIBSE TM14, and EULUMDAT. Apart from a few minor revisions, these data formats have remained essentially unchanged for the past several decades.

With the introduction of solid-state lighting with color-changing capabilities, there is a need to include spectral power distributions in these data representations. There is also a need to represent radiant and photosynthetically active radiation (PAR) intensity distributions for horticultural, aquacultural, and animal husbandry lighting applications. Unfortunately, it is difficult to impossible to incorporate such information in existing photometric data standards. It is also unreasonable to develop new standards specifically for specialized applications, such as horticultural lighting. This document therefore presents a standardized data format for use with all lighting applications.

The represented data may be obtained when testing or simulating the optical characteristics of a luminaire, which may then be used with lighting design and analysis software, architectural visualization software, or optical design software.

The data is formatted in accordance with W3C Extensible Markup Language (XML) 1.1 Recommendation and the W3C XML Schema Definition Language (XSD). This allows end users to utilize and view the data directly without the need for proprietary software.

In the event of any ambiguity or discrepancy with respect to the textual specification and the XML Schema in this document, the XML Schema shall take precedence.

In the event of any ambiguity or discrepancy with respect to the description of XML or XML Schema in this document, the W3C Recommendations shall take precedence.

1.2 Scope

This document specifies an electronic (XML-based) data format for the transfer of luminaire optical data useful for lighting design and analysis.

Details about the XML document format, XML schema, XSLT transforms, and more can be found at the W3C's website (see **Sections 2.15 and 2.16**), the authority for the XML document format. This document is intended as a description of a specific implementation of an XML document, and is not a tutorial on the XML document format itself.

2.0 Normative References

2.1 ANSI C78.377-2017

National Electrical Manufacturers Association. American National Standard for Electric Lamps – Specifications for the Chromaticity of Solid-State Lighting Products. Rosslyn, Va: NEMA; 2017.

2.2 ANSI/ASABE S640

American Society of Agricultural and Biological Engineers. Quantities and Units of Electromagnetic Radiation for Plants (Photosynthetic Organisms). St. Joseph, Mich.: ASABE; 2017.

2.3 ANSI/IES LS-1-22

Illuminating Engineering Society. Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2022. Online: <http://www.ies.org/standards/definitions/>. (Accessed 2022 Mar 9).

2.4 CIE 13.3-1995

International Commission on Illumination (CIE). Method of Measuring and Specifying Colour Rendering Properties of Light Sources. Vienna: CIE; 1995.

2.5 CIE 15:2004, 3rd ed.

International Commission on Illumination (CIE). Colorimetry, 3rd ed. Vienna: CIE; 2004.

2.6 CIE 63-1984

International Commission on Illumination (CIE). The Spectroradiometric Measurement of Light Sources. Vienna: CIE; 1984.

2.7 CIE 117-1995

International Commission on Illumination (CIE). Discomfort Glare in Interior Lighting. Vienna: CIE; 1995.

2.8 CIE 121-1996

International Commission on Illumination (CIE). The Photometry and Goniophotometry of Luminaires. Vienna: CIE; 1996.

2.9 CIE 224-2017

International Commission on Illumination (CIE). CIE 2017 Colour Fidelity Index for Accurate Scientific Use. Vienna: CIE; 2017.

2.10 ECMA-404

ECMA International. The JSON Data Interchange Syntax, 2nd ed. Geneva: ECMA International; 2017. Online: <https://www.ecma-international.org/>. (Accessed 2022 Mar 9).

2.11 IES LM-63-19

Illuminating Engineering Society. Standard File Format for Electronic Transfer of Photometric Data. New York: IES; 2019.

2.12 IES LM-75-19

Illuminating Engineering Society. Approved Method: Guide to Goniophotometer Measurements and Types and Photometric Coordinate Systems. New York: IES; 2019.

2.13 IES TM-27-20

Illuminating Engineering Society. Technical Memorandum: IES Standard Format for the Electronic Transfer of Spectral Data. New York: IES; 2020.

2.14 IES TM-30-20

Illuminating Engineering Society. Technical Memorandum: IES Method for Evaluating Light Source Color Rendition. New York: IES; 2020.

2.15 RFC 4122

Leach P, Mealling M, Salz R A. Universally Unique IDentifier (UUID) URN Namespace. Internet Engineering Task Force; 2015 Jul. (RFC 4122). Online: <https://datatracker.ietf.org/doc/html/rfc4122>. (Accessed 2022 Mar 9).

2.16 W3C Extensible Markup Language (XML) 1.1, 2nd ed.

(<http://www.w3.org/TR/xml11>). (Accessed 2022 Mar 9).

2.17 W3C XML Schema Definition Language (XSD) 1.1. Part 1: Structures

(<http://www.w3.org/TR/xmlschema11-1>). (Accessed 2022 Mar 9).

2.18 W3C XML Schema Definition Language (XSD) 1.1. Part 2: Datatypes

(<http://www.w3.org/TR/xmlschema11-2>). (Accessed 2022 Mar 9).

3.0 Definitions

3.1 bandpass correction

The correction of measured spectral data to account for the bandpass width of the detector element and the scanning interval. There are several methods of bandpass correction, and this correction is usually performed either in the measurement equipment itself or by the data processing software.

3.2 channel multiplier

A number between 0 and 1 that represents the contribution of a single-channel emitter to the measured or simulated spatial intensity distribution and flux of a multichannel luminaire. For example, if a three-channel theatrical luminaire with independently dimmable red, green, and blue light-emitting diode light (LED) emitters generates 1,000 lumens of white light with a correlated color temperature (CCT) of 6500 K, the corresponding channel multipliers might be 0.63, 0.74, and 1.00 respectively.

3.3 dimensions

For the purposes of this document, a luminaire shall be represented as a rectangular or cylindrical object with dimensions of length, width, and (optionally) height, as shown in **Figure 3-1**, where the width dimension is parallel to the 0 – 180 degrees axis of the intensity distribution, and the length dimension is parallel to the 90 – 270 degrees axis of the intensity distribution. (This convention conforms with CIE 121-1996; see **Section 2.8**.)

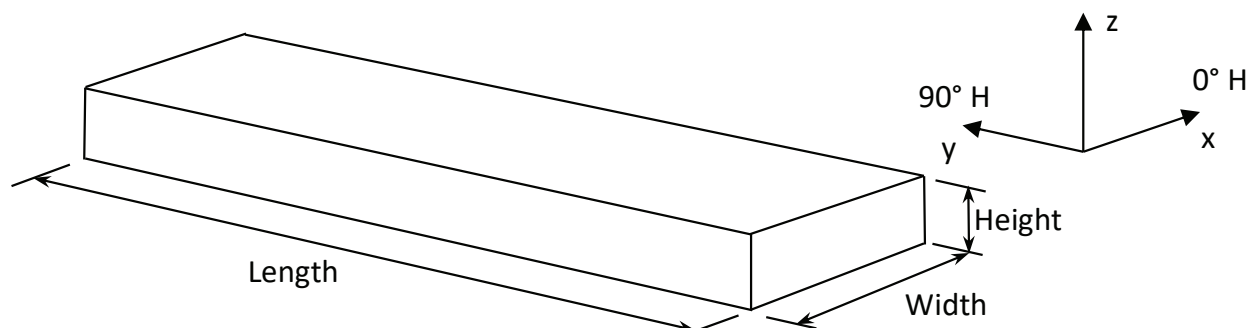


Figure 3-1. Luminaire dimensions.

Note: The purpose of this geometric object is to represent planar surfaces for the emission areas (see **Section 3.5**). It is not intended to represent the physical geometry of the luminaire housing.

The x-y-z coordinate system consists of the x-axis aligned with 0 degrees horizontal, the y-axis aligned with 90 degrees horizontal, and the z-axis aligned with zenith (+90 degrees vertical). The origin is coincident with the geometric center of the rectangular or cylindrical object (which may be different from the emitter center used for measurement purposes).

The four side faces of the enclosing box shall be identified as:

- C0 Perpendicular to 0-degree (+x) axis
- C90 Perpendicular to 90-degree (+y) axis
- C180 Perpendicular to 180-degree (-x) axis
- C270 Perpendicular to 270-degree (-y) axis

All dimensions shall be expressed in meters.

3.4 document

Instead of using file as the description for this data storage mechanism, the term document will be used. The term document is more appropriate when the information may not actually exist as a physical file, but may instead be generated dynamically and electronically transmitted.

3.5 emission areas

The areas of a specified face of the luminaire from which optical radiation is emitted. Multiple emission areas per face are allowed, including rectangular and elliptical shapes. They are intended primarily for use in CIE Unified Glare Rating (UGR) and similar visual glare calculations.

3.6 emitter

Any device that emits electromagnetic (“optical”) radiation within the wavelength range of 100 nm to 1,000 μm .

Note: A device that emits optical radiation within the wavelength range of 380 nm to 780 nm (that is, within the visible range) may also be referred to as a light source.

3.7 emitter center

The point that represents the origin for intensity distribution measurements. It may be offset from the geometric center of the rectangular or cylindrical luminaire representation. Also referred to as the “photometric center” for goniophotometric applications, it is the point in a luminaire or emitter from which the inverse-square law operates most closely in the direction of maximum intensity.

3.8 far-field versus near-field

Far-field intensity measurements assume a point emitter. In practice, the distance from the emitter to the sensor should be at least five times the largest diagonal dimension of the emission area (i.e., the “five-times” rule for goniophotometry).

Near-field illuminance, irradiance, photon flux density, and spectral irradiance measurements are typically made on a regular grid on a plane. No assumptions are made about the dimensions or geometry of the emitter(s).

3.9 geometric center

The center of the rectangular or cylindrical object defining the luminaire dimensions.

3.10 gonioradiometer

For the purpose of this document: an instrument for measuring the directional radiant intensity, spectral radiant intensity, or spectrally weighted radiant intensity distribution of an emitter. This definition encompasses goniophotometers, gonioradiometers, and goniospectroradiometers.

3.11 luminous

For the purpose of this document, the term *luminous* is synonymous with *photometric*.

3.12 photon flux

The rate of flow of photons. It is analogous to radiant flux, except that it is measured in micromoles of photons per second ($\mu\text{mol/s}$) rather than joules of energy per second (watts). (See **Annex B – Photon Flux** for further information.) Also known as *quantum flux*.

3.13 photon intensity

The photon flux emitted by an emitter in a given direction. It is measured in micromoles of photons per second per steradian ($\mu\text{mol/s}\cdot\text{sr}$). Analogous to radiant intensity.

3.14 photon flux density

The areal density of photon flux incident upon a real or imaginary surface. It is measured in micromoles of photons per second per square meter ($\mu\text{mol/sec}\cdot\text{m}^2$). Analogous to irradiance.

4.0 XML Schema for Luminaire Optical Data

The luminaire optical data document follows the W3C XML Schema presented in **Section 4.7**.

The XML root element shall be <IESTM33-22>.

The XML document is separated into six elements:

1. The required `Version` element, which identifies the XML schema version
2. The required `Header` element, which contains all generic information about the luminaire
3. The optional `Luminaire` element, which contains information about the luminaire dimensions, shape, and number of emitters
4. The optional `Equipment` element, which contains information about the measurement equipment used to perform the optical measurements
5. The required `Emitter` element, which contains information about the emitter(s), including generic information, intensity distribution data, and spectral power distribution data
6. The optional `CustomData` element, which contains information specific to a particular application; for example, government-mandated environmental criteria

4.1 Version Element

The required `Version` element identifies the XML schema version. It shall be "1.0".

4.2 Header Element

The required `Header` element is the parent of the document header section. This section contains information that is not specific to the luminaire data. Header elements are listed in **Table 1**.

Table 1. Header Element Fields

Element Description	Element Name	Data Type	Required	Document Section
Manufacturer	Manufacturer	xs:string	Optional	4.2.1
Catalog Number	CatalogNumber	xs:string	Optional	4.2.2
GTIN	GTIN	xs:integer	Optional	4.2.3
Description	Description	xs:string	Yes	4.2.4
Laboratory	Laboratory	xs:string	Yes	4.2.5
Report Number	ReportNumber	xs:string	Yes	4.2.6
Report Date	ReportDate	xs:date	Yes	4.2.7
Document Creator	DocumentCreator	xs:string	Optional	4.2.8
Document Creation Date	DocumentCreationDate	xs:date	Optional	4.2.9
Unique Identifier	UniqueIdentifier	xs:string	Yes	4.2.10
Comment	Comment	xs:string	Optional	4.2.11
Reference	Reference	xs:string	Optional	4.2.12
More Information URI	MoreInfoURI	xs:anyURI	Optional	4.2.13

4.2.1 Manufacturer Element The optional `Manufacturer` element identifies the manufacturer of the luminaire or emitter.

4.2.2 Catalog Number Element The optional `CatalogNumber` element identifies the manufacturer's product catalog number.

4.2.3 GTIN Element The optional `GTIN` element identifies the Global Trade Item Number (GTIN) of the luminaire or emitter.

4.2.4 Description Element The required `Description` element contains a text description of the luminaire or emitter.

4.2.5 Laboratory Element The required `Laboratory` element identifies the testing laboratory name that performed the data measurements. If the data was generated from a computer simulation and not tested at a laboratory, this field shall contain the name of the company that generated the data.

4.2.6 Report Number Element The required `ReportNumber` element identifies the testing laboratory report number.

4.2.7 Report Date Element The required `ReportDate` element identifies the testing laboratory report date using the XSD Date data type, YYYY-MM-DD.

4.2.8 Document Creator Element The optional `DocumentCreator` element identifies the creator of the document, which may be a test lab, a research group, a standard body, a company, or an individual.

4.2.9 Document Creation Date Element The optional `DocumentCreationDate` element identifies the document creation date using the XSD Date data type, YYYY-MM-DD.

4.2.10 Unique Identifier Element The required `UniqueIdentifier` element contains a Universally Unique Identifier (UUID) as defined by RFC 4122 (see **Section 2.15**). Most scientific programming language libraries include functions that will automatically generate UUIDs.

A UUID is a unique 128-bit value expressed as a string with 32 hexadecimal digits in the format xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx. For example:

21EC2020-3AEA-4069-A2DD-08002B30309D

The intent of the UUID is to uniquely identify the document's measurement data. The document creator shall generate a new UUID whenever the document's measurement data is modified.

The document creator shall also generate a new UUID whenever an existing photometric data file is converted to this standard.

4.2.11 Comment Element The optional `Comment` element provides arbitrary additional information relating to the tested and reported data.

Multiple `Comment` elements are allowed.

4.2.12 Reference Element The optional `Reference` element identifies one or more regulatory standards or other documents with which the luminaire or emitter complies, or the regulatory standards or other documents to which the laboratory measurements conform. This may, for example, be a document name or a web address.

Multiple `Reference` elements are allowed.

4.2.13 More Information URI Element The optional `MoreInfoURI` element provides a Uniform Resource Identifier (URI) address for further luminaire information.

4.3 Luminaire Element

The optional `Luminaire` element is the parent of the luminaire data. Elements are detailed in **Table 2**.

Table 2. Luminaire Fields

Element Description	Element Name	Data Type	Required	Document Section
Dimensions	<code>Dimensions</code>	XML Element	Yes	4.3.1
Shape	<code>Shape</code>	<code>xs:string</code>	Optional	4.3.2
Number of Emitters	<code>NumEmitter</code>	<code>xs:int</code>	Yes	4.3.3

4.3.1 Dimensions Element The required `Dimensions` element is the parent of the luminaire dimensions data. Elements are detailed in **Table 3**.

Table 3. Dimensions Fields

Element Description	Element Name	Data Type	Required	Document Section
Length	<code>Length</code>	<code>xs:decimal</code>	Yes	4.3.1.1
Width	<code>Width</code>	<code>xs:decimal</code>	Yes	4.3.1.2
Height	<code>Height</code>	<code>xs:decimal</code>	Yes	4.3.1.3

4.3.1.1 Length Element The required `Length` element specifies the length of the luminaire housing in meters.

4.3.1.2 Width Element The required `Width` element specifies the width of the luminaire housing in meters.

4.3.1.3 Height Element The required `Height` element specifies the height of the luminaire housing in meters. (To support legacy photometric data file formats, the value of this element may be zero.)

4.3.2 Shape Element The luminaire may optionally be represented as an axis-aligned cylindrical object. The optional `Shape` element identifies the aligned axis (see **Figure 2**). Valid values are listed in **Table 4**.

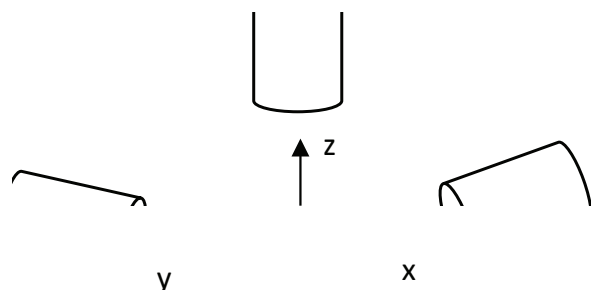


Figure 4-2. Axis-aligned cylindrical luminaire orientations.

Table 4: Shape Types

Type	Description
<code>Align_X</code>	Luminaire is circular in the y-z plane
<code>Align_Y</code>	Luminaire is circular in the x-z plane
<code>Align_Z</code>	Luminaire is circular in the x-y plane

4.3.3 Number of Emitters Element The required `NumEmitter` element specifies the number of Emitter elements. (See **Section 4.5** for details.)

The Number of Emitters will typically be defined by the number of unique sets of output data that this file should contain.

For an RGBW LED it may make the most sense to use a single Emitter with 4 child `EmitterSpectral` elements since only the spectra changes with each LED, but for a luminaire with cool white and warm white PCBs, it may be more sense to use two Emitter elements, each with a single child `EmitterSpectral` element since other items like the Intensity Data or Irradiance data will change as well. See Informative Appendix E for additional information.

4.4 Equipment Element

The optional `Equipment` element is the parent of measurement equipment data. Elements are detailed in **Table 5**.

Table 5. Equipment Field

Element Description	Element Name	Data Type	Required	Document Section
Gonioradiometer	<code>Gonioradiometer</code>	XML Element	Optional	4.4.1
Integrating Sphere	<code>IntegratingSphere</code>	XML Element	Optional	4.4.2
Spectroradiometer	<code>Spectroradiometer</code>	XML Element	Optional	4.4.3

4.4.1 Gonioradiometer Element The optional `Gonioradiometer` element is the parent of gonioradiometric data. This element contains information that is specific to the gonioradiometer data. Elements are detailed in **Table 6**.

Table 6. Gonioradiometer Fields

Element Description	Element Name	Data Type	Required	Document Section
Type	<code>Type</code>	xs:string	Yes	4.4.1.1
Measurement Equipment	<code>MeasurementEquipment</code>	xs:string	Optional	4.4.1.2

4.4.1.1 Type Element The required `Type` element identifies the gonioradiometer type. Valid values are listed in **Table 7**.

Table 7. Gonioradiometer Types

Type	Reference
CIE _ A	CIE 121-1996
CIE _ B	CIE 121-1996
CIE _ C	CIE 121-1996
IES _ A	IES LM-75-19
IES _ B	IES LM-75-19
IES _ C	IES LM-75-19
CUSTOM	Not applicable

4.4.1.2 Measurement Equipment Element The optional `MeasurementEquipment` element contains a description of the equipment used to measure the gonioradiometric data.

Multiple `MeasurementEquipment` elements are allowed.

4.4.2 Integrating Sphere Element The optional `IntegratingSphere` element is the parent of integrating sphere data. Elements are detailed in **Table 8**.

Table 8: Integrating Sphere Fields

Element Description	Element Name	Data Type	Required	Document Section
Measurement Equipment	<code>MeasurementEquipment</code>	xs:string	Yes	4.4.2.1

4.4.2.1 Measurement Equipment Element The required MeasurementEquipment element contains a description of the equipment used to measure the integrating sphere data.

Multiple MeasurementEquipment elements are allowed.

4.4.3 Spectroradiometer Element The optional Spectroradiometer element is the parent of spectroradiometer data. Elements are detailed in **Table 9**.

Table 9. Spectroradiometer Fields

Element Description	Element Name	Data Type	Required	Document Section
Measurement Equipment	MeasurementEquipment	xs:string	Yes	4.4.3.1
Bandwidth FWHM	BandwidthFWHM	xs:decimal	Optional	4.4.3.2
Bandwidth Corrected	BandwidthCorrected	xs:boolean	Optional	4.4.3.3
Bandwidth Method	BandwidthMethod	xs:string	Optional	4.4.3.4

4.4.3.1 Measurement Equipment Element The required MeasurementEquipment element contains a description of the equipment used to measure the spectral data.

4.4.3.2 Bandwidth FWHM Element The optional BandwidthFWHM attribute specifies the spectroradiometer full-width half-maximum bandwidth, in nanometers.

4.4.3.3 Bandwidth Corrected Element The optional BandwidthCorrected attribute specifies that bandwidth correction has been applied to the measured data.

4.4.3.4 Bandwidth Method Element The optional BandwidthMethod attribute documents the bandwidth correction method.

4.5 Emitter Element

The required Emitter element is the parent of emitter data. Elements are detailed in **Table 10**.

Multiple Emitter elements are allowed as per NumEmitter element (**Section 4.3.3**).

Table 10. Light Source Fields

Element Description	Element Name	Data Type	Required	Document Section
Quantity	Quantity	xs:int	Yes	4.5.1
Description	Description	xs:string	Yes	4.5.2
Catalog Number	CatalogNumber	xs:string	Optional	4.5.3
Rated Lumens	RatedLumens	xs:decimal	Optional	4.5.4
Input Wattage	InputWattage	xs:decimal	Yes	4.5.5
Power Factor	PowerFactor	xs:decimal	Optional	4.5.6
Ballast Factor	BallastFactor	xs:decimal	Optional	4.5.7
Tilt Angles	TiltAngles	XML Element	Optional	4.5.8
Correlated Color Temperature (CCT)	ColorTemperature	XML Element	Optional	4.5.9
Color Rendering	ColorRendering	XML Element	Optional	4.5.10
Duv	Duv	xs:decimal	Optional	4.5.11
S/P Ratio	SPRatio	xs:decimal	Optional	4.5.12

(continued on next page)

Table 10. Light Source Fields *(continued)*

Element Description	Element Name	Data Type	Required	Document Section
Data Generation	DataGeneration	XML Element	Optional	4.5.13
Luminous Data	LuminousData	XML Element	Optional	4.5.14
Radiant Data	RadiantData	XML Element	Optional	4.5.15
Photon Data	PhotonData	XML Element	Optional	4.5.16
Spectral Data	SpectralData	XML Element	Optional	4.5.17
Angular Color Data	AngularColor	XML Element	Optional	4.5.18
Illuminance Data	IllumData	XML Element	Optional	4.5.19
Irradiance Data	IrradData	XML Element	Optional	4.5.20
Photon Flux Density Data	PFDDData	XML Element	Optional	4.5.21
Spectral Irradiance Data	SpecIrradData	XML Element	Optional	4.5.22
Channels	Channels	XML Element	Optional	4.5.23
Emission Areas	EmissionAreas	XML Element	Optional	4.5.24
Emitter Center	EmitterCenter	XML Element	Optional	4.5.25
Regulatory	Regulatory	XML Element	Optional	4.2.26

Note: The `Emitter` element shall include at least one of the following elements: `LuminousData`, `RadiantData`, `PhotonData`, `SpectralData`, `AngularColor`, `IllumData`, `IrradData`, `PFDDData`, and `SpecIrradData`.

4.5.1 Quantity Element The required `Quantity` element specifies the number of sources within this emitter. This could be the number of individual LEDs within this emitter, or 1 if there is a single source within this Emitter or the number of sources is not being disclosed.

4.5.2 Description Element The required `Description` element describes the emitter type.

4.5.3 Catalog Number Element The optional `CatalogNumber` element specifies the emitter manufacturer's catalog number.

4.5.4 Rated Emitter Lumens Element The optional `RatedLumens` element specifies the manufacturer's rated emitter lumens. For variable-CCT emitters, this value shall be the maximum rated emitter lumens.

4.5.5 Input Wattage Element The required `InputWattage` element specifies the input wattage.

4.5.6 Power Factor Element The optional `PowerFactor` element specifies the power factor.

4.5.7 Ballast Factor Element The optional `BallastFactor` element specifies the ratio of the flux emitted by the emitter when operated with a commercial ballast to the flux emitted by the same emitter when operated with a standard (reference) ballast used for rating lamp lumens. For application purposes, the ballast factor is used to adjust the emitter performance data from laboratory test conditions to actual field conditions.

4.5.8 Tilt Angles Element The optional `TiltAngles` element is the parent of the emitter tilt angle data. Elements are detailed in **Table 11**.

Table 11. Tilt Angle Fields

Element Description	Element Name	Data Type	Required	Document Section
Number of Angles	NumberAngles	xs:int	Yes	4.5.8.1
Tilt	Tilt	XML Element	Yes	4.5.8.2

4.5.8.1 Number of Angles Element The required `NumberAngles` element specifies the number of tilt angles.

4.5.8.2 Tilt Element The required `Tilt` element specifies the emitter intensity multiplier for the specified tilt angle (in degrees). For example:

```
<Tilt angle="30.0">0.87</Tilt>
```

4.5.9 Correlated Color Temperature Element The optional `ColorTemperature` element is the parent of the correlated color temperature (CCT) range data. Elements are detailed in **Table 12**.

Table 12. Correlated Color Temperature Range Fields

Element Description	Element Name	Data Type	Required	Document Section
Fixed CCT	<code>FixedCCT</code>	xs:int	Optional	4.5.9.1
Minimum CCT	<code>MinCCT</code>	xs:int	Optional	4.5.9.2
Maximum CCT	<code>MaxCCT</code>	xs:int	Optional	4.5.9.3

4.5.9.1 Fixed CCT Element The optional `FixedCCT` element specifies the correlated color temperature of a fixed-CCT emitter.

4.5.9.2 Minimum CCT Element The optional `MinCCT` element specifies the minimum correlated color temperature of a variable-CCT emitter.

4.5.9.3 Maximum CCT Element The optional `MaxCCT` element specifies the maximum correlated color temperature of a variable-CCT emitter.

4.5.10 Color Rendering Element The optional `ColorRendering` element is the parent of color rendering metrics data. Elements are detailed in **Table 13**.

Table 13. Color Rendering Metrics Fields

Element Description	Element Name	Data Type	Required	Document Section
CIE CRI	<code>CIE_CRI</code>	XML Element	Optional	4.5.10.1
IES TM-30	<code>IES_TM30</code>	XML Element	Optional	4.5.10.2

4.5.10.1 CIE CRI Element The optional `CIE_CRI` element is the parent of CIE Colour Rendering Indices data. Elements are detailed in **Table 14**.

Table 14. CIE CRI Fields

Element Description	Element Name	Data Type	Required	Document Section
R_a	<code>Ra</code>	xs:int	Yes	4.5.10.1.1
R_9	<code>R9</code>	xs:int	Optional	4.5.10.1.2

4.5.10.1.1 R_a Element. The required `Ra` element specifies the CIE General Colour Rendering Index, R_a .

4.5.10.1.2 R_9 Element. The optional `R9` element specifies the CIE Special Colour Rendering Index R_9 .

4.5.10.2 IES_TM-30 Element The optional IES _TM30 element is the parent of IES TM-30 color rendering metrics data. Elements are detailed in **Table 15**.

Table 15: IES TM-30 Fields

Element Description	Element Name	Data Type	Required	Document Section
R_f	Rf	xs:int	Yes	4.5.10.2.1
R_g	Rg	xs:int	Yes	4.5.10.2.2
R_{fh01}	Rfh01	xs:int	Optional	4.5.10.2.3
R_{fh02}	Rfh02	xs:int	Optional	4.5.10.2.3
R_{fh03}	Rfh03	xs:int	Optional	4.5.10.2.3
R_{fh04}	Rfh04	xs:int	Optional	4.5.10.2.3
R_{fh05}	Rfh05	xs:int	Optional	4.5.10.2.3
R_{fh06}	Rfh06	xs:int	Optional	4.5.10.2.3
R_{fh07}	Rfh07	xs:int	Optional	4.5.10.2.3
R_{fh08}	Rfh08	xs:int	Optional	4.5.10.2.3
R_{fh09}	Rfh09	xs:int	Optional	4.5.10.2.3
R_{fh10}	Rfh10	xs:int	Optional	4.5.10.2.3
R_{fh11}	Rfh11	xs:int	Optional	4.5.10.2.3
R_{fh12}	Rfh12	xs:int	Optional	4.5.10.2.3
R_{fh13}	Rfh13	xs:int	Optional	4.5.10.2.3
R_{fh14}	Rfh14	xs:int	Optional	4.5.10.2.3
R_{fh15}	Rfh15	xs:int	Optional	4.5.10.2.3
R_{fh16}	Rfh16	xs:int	Optional	4.5.10.2.3
$R_{cs,h01}$	Rcsh01	xs:int	Optional	4.5.10.2.4
$R_{cs,h02}$	Rcsh02	xs:int	Optional	4.5.10.2.4
$R_{cs,h03}$	Rcsh03	xs:int	Optional	4.5.10.2.4
$R_{cs,h04}$	Rcsh04	xs:int	Optional	4.5.10.2.4
$R_{cs,h05}$	Rcsh05	xs:int	Optional	4.5.10.2.4
$R_{cs,h06}$	Rcsh06	xs:int	Optional	4.5.10.2.4
$R_{cs,h07}$	Rcsh07	xs:int	Optional	4.5.10.2.4
$R_{cs,h08}$	Rcsh08	xs:int	Optional	4.5.10.2.4
$R_{cs,h09}$	Rcsh09	xs:int	Optional	4.5.10.2.4
$R_{cs,h10}$	Rcsh10	xs:int	Optional	4.5.10.2.4
$R_{cs,h11}$	Rcsh11	xs:int	Optional	4.5.10.2.4
$R_{cs,h12}$	Rcsh12	xs:int	Optional	4.5.10.2.4
$R_{cs,h13}$	Rcsh13	xs:int	Optional	4.5.10.2.4
$R_{cs,h14}$	Rcsh14	xs:int	Optional	4.5.10.2.4
$R_{cs,h15}$	Rcsh15	xs:int	Optional	4.5.10.2.4
$R_{cs,h16}$	Rcsh16	xs:int	Optional	4.5.10.2.4

4.5.10.2.1 R_f Element. The required Rf element specifies the IES TM-30 fidelity index, R_f .

4.5.10.2.2 R_g Element. The required Rg element specifies the IES TM-30 gamut index, R_g .

4.5.10.2.3 R_{fh}^* Elements. The optional Rfh* elements specify the sixteen local fidelity values within each hue-angle range.

4.5.10.2.4 $R_{cs,h}^*$ Elements. The optional Rcsh* elements specify the sixteen local chroma shift values within each hue-angle range.

4.5.11 Duv Element The optional `Duv` element specifies the closest distance of the emitter chromaticity to the Planckian locus, as defined in ANSI C78.377-2017.

4.5.12 S/P Ratio Element The optional `SPRatio` element specifies the light source scotopic-to-photopic lumens ratio.

4.5.13 Data Generation Element The optional `DataGeneration` element is the parent of intensity data generation information. Elements are detailed in **Table 16**.

Table 16. Data Generation Fields

Element Description	Element Name	Data Type	Required	Document Section
Simulation	<code>Simulation</code>	xs:boolean	Optional	4.5.13.1
Laboratory	<code>Laboratory</code>	XML Element	Optional	4.5.13.2
Intensity Scaling	<code>IntensityScaling</code>	xs:boolean	Optional	4.5.13.3
Angle Interpolation	<code>AngleInterpolation</code>	xs:boolean	Optional	4.5.13.4

4.5.13.1 Simulation Element The optional `Simulation` element, if “true”, indicates that the intensity data were generated by means of a computer simulation rather than measured.

4.5.13.2 Laboratory Element The optional `Laboratory` element is the parent of the Laboratory Certification data. Elements are detailed in **Table 17**.

Table 17: Laboratory Fields

Element Description	Element Name	Data Type	Required	Document Section
Type	<code>Certification</code>	xs:string	Yes	4.5.13.2.1
Approval Body	<code>ApprovalBody</code>	xs:string	Yes	4.5.13.2.2
Approval Scope	<code>ApprovalScope</code>	xs:string	Yes	4.5.13.2.3
Measurement Uncertainty	<code>MeasUncertainty</code>	XML Element	Yes	4.5.13.2.4

4.5.13.2.1 Certification Element The required `Certification` element identifies the certification type. Valid values are listed in **Table 18**.

Table 18: Certification Types

Type	Element Description
Accredited	Accredited test laboratory
Associated	Associated test laboratory
Customer	Customer test facilities
None	–

4.5.13.2.2 Approval Body Element The required `ApprovalBody` element identifies the approval body. For example, “NVLAP”.

4.5.13.2.3 Approval Scope Element The required `ApprovalScope` element identifies the scope of the approval. For example, “IES LM-79 [Sections 9, 10, 12]”.

4.5.13.2.4 Measurement Uncertainty Element The required `MeasUncertainty` element is the parent of the Measurement Uncertainty data. Elements are detailed in **Table 19**.

Multiple `MeasUncertainty` elements are allowed.

Table 19: Measurement Uncertainty Fields

Element Description	Element Name	Data Type	Required	Document Section
Measurement Type	<code>MeasurementType</code>	xs:string	Yes	4.5.13.2.4.1
Uncertainty	<code>Uncertainty</code>	xs:decimal	Yes	4.5.13.2.4.2

4.5.13.2.4.1 Measurement Type Element The required `MeasurementType` element identifies the measurement type.

4.5.13.2.4.2 Uncertainty Element The required `Uncertainty` element specifies the measurement uncertainty.

4.5.13.3 Intensity Scaling Element The optional `IntensityScaling` element, if “true”, indicates that the reported intensity data has been uniformly scaled with respect to the laboratory measurements. A required attribute titled “comment” shall be used to describe the Intensity Scaling. Examples of the comment attribute can be found in Annex C.

The `IntensityScaling` boolean is informative and does not change or alter the interpretation of the Intensity values in the file. It informs the file reader that the `IntData` values have previously been scaled in some way. Examples of applications of `IntensityScaling` can be found in Annex C.

4.5.13.4 Angle Interpolation Element The optional `AngleInterpolation` element, if “true”, indicates that the reported intensity data at the reported horizontal and/or vertical angles have been interpolated with respect to the laboratory measurements.

4.5.14 Luminous Data Element The optional `LuminousData` element is the parent of luminous data. This element contains information that is specific to the luminous data. Elements are detailed in **Table 20**.

Table 20. Luminous Data Fields

Element Description	Element Name	Data Type	Required	Document Section
Luminous Intensity	<code>LuminousIntensity</code>	XML Element	Yes	4.5.14.1
Luminous Flux	<code>LuminousFlux</code>	<code>xs:decimal</code>	Optional	4.5.14.2

4.5.14.1 Luminous Intensity Element The required `LuminousIntensity` element is the parent of the luminous intensity data. Elements are detailed in **Table 21**.

Table 21. Luminous Intensity Data Fields

Element Description	Element Name	Data Type	Required	Document Section
Absolute Photometry	<code>AbsolutePhotometry</code>	<code>xs:boolean</code>	Yes	4.5.14.1.1
Symmetry	<code>Symm</code>	<code>xs:string</code>	Optional	4.5.14.1.2
Multiplier	<code>Multiplier</code>	<code>xs:decimal</code>	Optional	4.5.14.1.3
Number Measured	<code>NumberMeasured</code>	<code>xs:int</code>	Yes	4.5.14.1.4
Number of Horiz. Angles	<code>NumberHorz</code>	<code>xs:int</code>	Yes	4.5.14.1.5
Number of Vert. Angles	<code>NumberVert</code>	<code>xs:int</code>	Yes	4.5.14.1.6
Intensity Data	<code>IntData</code>	XML Element	Yes	4.5.14.1.7

4.5.14.1.1 Absolute Photometry Element The required `AbsolutePhotometry` element, if “true”, indicates that the luminous intensity measurements were performed using absolute photometry.

4.5.14.1.2 Symmetry Element The optional `Symm` element identifies the horizontal symmetry type. Valid values for CIE _ C and IES _ C gonioradiometer types only are shown in **Table 22**.

Table 22. Symmetry Types

Type	Description	Typical Horizontal Angle Range
Symm _ None	No symmetry about the vertical axis	0-360
Symm _ Bi _ 0	Bilateral symmetry about the vertical plane thru Horizontal 0-180 degree angles.	0-180
Symm _ Bi _ 90	Bilateral symmetry about the vertical plane thru Horizontal 90-270 degree angles	90-270
Symm _ Quad	Quadrilateral symmetry about the vertical planes thru Horizontal 0-180 degree angles Horizontal 90-270 degrees	0-90
Symm_Full	Symmetric in all vertical planes	0
Symm_Arbitrary	No horizontal or vertical symmetry	Undefined

If the Symm element is not present, the default value is Symm_None.

Note 1: Symm _ Arbitrary indicates that the intensity data represent measurements at arbitrary horizontal and vertical angles, such as for a geodesic sphere, an example of which is shown in Annex D. The angles are identified in the IntData element using the h and v attributes for horizontal and vertical angle.

Note 2: If the Symm element is Symm _ Arbitrary, the values of the NumberHorz and NumberVert elements shall be zero.

Note 3: Typical horizontal angles are the minimum range based on the indicated symmetry. The full data set can be included, but software reading these files may choose to present a warning indicating a potential mismatch of the indicated symmetry type and the IntData horizontal angles included in the file.

4.5.14.1.3 Multiplier Element The optional Multiplier element, if present, indicates a floating-point multiplier that shall be applied to the luminous intensity measurements. An required attribute titled "comment" shall be used to describe the Multiplier. Examples of the comment attribute can be found in Annex C.

If the Multiplier element is not present, the default value is 1.0.

4.5.14.1.4 Number Measured Element The required NumberMeasured element specifies the number of luminous intensity measurements.

4.5.14.1.5 Number of Horiz. Angles Element The required NumberHorz element specifies the number of horizontal angles. The value shall be zero if the measurement coordinate system does not consist of a set of planes.

4.5.14.1.6 Number of Vert. Angles Element The required NumberVert element specifies the number of vertical angles. The value shall be zero if the measurement coordinate system does not consist of a set of planes.

4.5.14.1.7 Intensity Data Element The required IntData element, when multiplied by the Multiplier element, specifies the luminous intensity data (in candelas) for the specified horizontal and vertical angles (in degrees). The element contains the value to be reported, and there are attributes for the horizontal and vertical angles at which this value occurs. For example:

```
<IntData h="0.0" v="65.0">44</IntData>
```

Note 1: Scientific notation (e.g., 1.23E-2) is allowed for the luminous intensity value.

Note 2: The `Multiplier` element enables luminous intensity values to be reported in candelas per kilolumen.

Multiple `IntData` elements are allowed.

4.5.14.2 Luminous Flux Element The optional `LuminousFlux` element specifies the luminous flux in lumens.

4.5.15 Radiant Data Element The optional `RadiantData` element is the parent of radiant data. Elements are detailed in **Table 23**.

Table 23. Radiant Data Fields

Element Description	Element Name	Data Type	Required	Document Section
Minimum Wavelength	<code>MinWavelength</code>	xs:decimal	Yes	4.5.15.1
Maximum Wavelength	<code>MaxWavelength</code>	xs:decimal	Yes	4.5.15.2
Radiant Intensity	<code>RadiantIntensity</code>	XML Element	Yes	4.5.15.3
Radiant Flux	<code>RadiantFlux</code>	xs:decimal	Optional	4.5.15.4

4.5.15.1 Minimum Wavelength Element The required `MinWavelength` element specifies the minimum wavelength of the radiant intensity measurements, in nanometers.

4.5.15.2 Maximum Wavelength Element The required `MaxWavelength` element specifies the maximum wavelength of the radiant intensity measurements, in nanometers.

4.5.15.3 Radiant Intensity Element The required `RadiantIntensity` element is the parent of the radiant intensity data. Elements are detailed in **Table 24**.

Table 24. Radiant Intensity Data Fields

Element Description	Element Name	Data Type	Required	Document Section
Absolute	<code>Absolute</code>	xs:boolean	Optional	4.5.15.3.1
Symmetry	<code>Symm</code>	xs:string	Optional	4.5.15.3.2
Multiplier	<code>Multiplier</code>	xs:decimal	Optional	4.5.15.3.3
Number Measured	<code>NumberMeasured</code>	xs:int	Yes	4.5.15.3.4
Number of Horiz. Angles	<code>NumberHorz</code>	xs:int	Yes	4.5.15.3.5
Number of Vert. Angles	<code>NumberVert</code>	xs:int	Yes	4.5.15.3.6
Intensity Data	<code>IntData</code>	XML Element	Yes	4.5.15.3.7

4.5.15.3.1 Absolute Element The optional `Absolute` element, if “true”, indicates that the radiant intensity measurements were performed using absolute radiometry.

If the `Absolute` element is not present, the default value is “true”.

4.5.15.3.2 Symmetry Element The optional `Symm` element identifies the horizontal symmetry type. Valid values for CIE _ C and IES _ C goniometer types only are shown in **Table 25**.

Table 25. Symmetry Types

Type	Description	Typical Horizontal Angle Range
Symm _ None	No symmetry about the vertical axis	0-360
Symm _ Bi _ 0	Bilateral symmetry about the vertical plane thru Horizontal 0-180 degree angles	0-180
Symm _ Bi _ 90	Bilateral symmetry about the vertical plane thru Horizontal 90-270 degree angles	90-270
Symm _ Quad	Quadrilateral symmetry about the vertical planes thru Horizontal 0-180 degree angles Horizontal 90-270 degree angles	0-90
Symm _ Full	Axial symmetry, identical vertical planes for every Horizontal angle	0
Symm _ Arbitrary	No symmetry, not a regular grid of horizontal and vertical angles	Undefined

If the Symm element is not present, the default value is Symm _ None.

Note 1: Symm _ Arbitrary indicates that the intensity data represent measurements at arbitrary horizontal and vertical angles, such as for a geodesic sphere, an example of which is shown in Annex D.

Note 2: If the Symm element is Symm _ Arbitrary, the values of the NumberHorz and NumberVert elements shall be zero.

Note 3: Typical horizontal angles are the minimum range based on the indicated symmetry. The full data set can be included, but software reading these files may choose to present a warning indicating a potential mismatch of the indicated symmetry type and the IntData horizontal angles included in the file.

4.5.15.3.3 Multiplier Element The optional Multiplier element, if present, indicates a floating-point multiplier that shall be applied to the radiant intensity measurements. A required attribute titled "comment" shall be used to describe the Multiplier. Examples of the comment attribute can be found in Annex C.

4.5.15.3.4 Number Measured Element The required NumberMeasured element specifies the number of radiant intensity measurements.

4.5.15.3.5 Number of Horiz. Angles Element The required NumberHorz element specifies the number of horizontal angles. The value shall be zero if the measurement coordinate system does not consist of a set of planes.

4.5.15.3.6 Number of Vert. Angles Element The required NumberVert element specifies the number of vertical angles. The value shall be zero if the measurement coordinate system does not consist of a set of planes. The angles are identified in the IntData element using the h and v attributes for horizontal and vertical angle.

4.5.15.3.7 Intensity Data Element The required IntData element, when multiplied by the Multiplier element, specifies the radiant intensity data (in watts per steradian) for the specified horizontal and vertical angles (in degrees). The element contains the value to be reported, and there are attributes for the horizontal and vertical angles at which this value occurs. For example:

```
<IntData h="0.0" v="65.0">44</IntData>
```

Note 1: Scientific notation (e.g., 1.23E-2) is allowed for the radiant intensity value.

Note 2: The `Multiplier` element enables radiant intensity values to be reported in microwatts per steradian or milliwatts per steradian.

Multiple `IntData` elements are allowed.

4.5.15.3.8 Radiant Flux Element The optional `RadiantFlux` element specifies the radiant flux in watts.

4.5.16 Photon Data Element The optional `PhotonData` element is the parent of photon data. Elements are detailed in **Table 26**.

Table 26. Photon Data Fields

Element Description	Element Name	Data Type	Required	Document Section
Minimum Wavelength	<code>MinWavelength</code>	xs:decimal	Yes	4.5.16.1
Maximum Wavelength	<code>MaxWavelength</code>	xs:decimal	Yes	4.5.16.2
Photon Intensity	<code>PhotonIntensity</code>	XML Element	Yes	4.5.16.3
Photon Flux	<code>PhotonFlux</code>	xs:decimal	Optional	4.5.16.4

4.5.16.1 Minimum Wavelength Element The required `MinWavelength` element specifies the minimum wavelength of the photon measurements, in nanometers.

4.5.16.2 Maximum Wavelength Element The required `MaxWavelength` element specifies the maximum wavelength of the photon measurements, in nanometers.

4.5.16.3 Photon Intensity Element The required `PhotonIntensity` element is the parent of the photon intensity data. Elements are detailed in **Table 27**.

Table 27. Photon Intensity Data Fields

Element Description	Element Name	Data Type	Required	Document Section
Absolute	<code>Absolute</code>	xs:boolean	Optional	4.5.16.3.1
Symmetry	<code>Symm</code>	xs:string	Optional	4.5.16.3.2
Multiplier	<code>Multiplier</code>	xs:decimal	Optional	4.5.16.3.3
Number Measured	<code>NumberMeasured</code>	xs:int	Yes	4.5.16.3.4
Number of Horiz. Angles	<code>NumberHorz</code>	xs:int	Yes	4.5.16.3.5
Number of Vert. Angles	<code>NumberVert</code>	xs:int	Yes	4.5.16.3.6
Intensity Data	<code>IntData</code>	XML Element	Yes	4.5.16.3.7

4.5.16.3.1 Absolute Element The optional `Absolute` element, if “true”, indicates that the photon intensity measurements were performed using absolute radiometry.

If the `Absolute` element is not present, the default value is “true”.

4.5.16.3.2 Symmetry Element The optional `Symm` element identifies the horizontal symmetry type. Value values for CIE _ C and IES _ C goniometer types only are shown in **Table 28**.

Table 28. Symmetry Types

Type	Description	Typical Horizontal Angle Range
Symm _ None	No symmetry about the vertical axis	0-360
Symm _ Bi _ 0	Bilateral symmetry about the vertical plane thru Horizontal 0-180 degree angles	0-180
Symm _ Bi _ 90	Bilateral symmetry about the vertical plane thru Horizontal 90-270 degree angles	90-270
Symm _ Quad	Quadrilateral symmetry about the vertical planes thru Horizontal 0-180 degree angles Horizontal 90-270 degree angles	0-90
Symm _ Full	Axial symmetry, identical vertical planes for every Horizontal angle	0
Symm _ Arbitrary	Symm_Arbitrary - No symmetry, not a regular grid of horizontal and vertical angles	Undefined

If the Symm element is not present, the default value is Symm _ None.

Note 1: Symm _ Arbitrary indicates that the intensity data represent measurements at arbitrary horizontal and vertical angles, such as for a geodesic sphere, an example of which is shown in Annex D. The angles are identified in the IntData element using the h and v attributes for horizontal and vertical angle.

Note 2: If the Symm element is Symm _ Arbitrary, the values of the NumberHorz and NumberVert elements shall be zero.

Note 3: Typical horizontal angles are the minimum range based on the indicated symmetry. The full data set can be included, but software reading these files may choose to present a warning indicating a potential mismatch of the indicated symmetry type and the IntData horizontal angles included in the file.

4.5.16.3.3 Multiplier Element The optional Multiplier element, if present, indicates a floating-point multiplier that shall be applied to the photon intensity measurements. A required attribute titled "comment" shall be used to describe the Multiplier. Examples of the comment attribute can be found in Annex C.

If the Multiplier element is not present, the default value is 1.0.

4.5.16.3.4 Number Measured Element The required NumberMeasured element specifies the number of photon intensity measurements.

4.5.16.3.5 Number of Horiz. Angles Element The required NumberHorz element specifies the number of horizontal angles. The value shall be zero if the measurement coordinate system does not consist of a set of planes.

4.5.16.3.6 Number of Vert. Angles Element The required NumberVert element specifies the number of vertical angles. The value shall be zero if the measurement coordinate system does not consist of a set of planes.

4.5.16.3.7 Intensity Data Element The required IntData element, when multiplied by the Multiplier element, specifies the photon intensity data (in micromoles per steradian per second) for the specified horizontal and vertical angles (in degrees). The element contains the value to be reported, and there are attributes for the horizontal and vertical angles at which this value occurs. For example:

```
<IntData h="0.0" v="65.0">44</IntData>
```

Note: Scientific notation (e.g., 1.23E-2) is allowed for the photon intensity value.

Multiple IntData elements are allowed.

4.5.16.4 Photon Flux Element The optional `PhotonFlux` element specifies the photon flux in micromoles per second ($\mu\text{mol/s}$).

4.5.17 Spectral Data Element

The optional `SpectralData` element is the parent of spectroradiometric data. Elements are detailed in **Table 29**.

Table 29. Spectral Data Fields

Element Description	Element Name	Data Type	Required	Document Section
Spectral Intensity	<code>EmitterSpectral</code>	XML Element	Optional	4.5.17.1
Angular Spectral	<code>AngularSpectral</code>	XML Element	Optional	4.5.17.2

4.5.17.1 Emitter Spectral Element

The optional `EmitterSpectral` element is the parent of emitter spectral data. Elements are detailed in **Table 30**.

Multiple `EmitterSpectral` elements are allowed. If the same emitter is capable of emitting multiple spectra, each spectrum can be represented as an individual `EmitterSpectral` element. For example, luminaires with RGBW LED chips are capable of emitting multiple spectra using the same array of chips. See Annex E for additional information.

Table 30. Emitter Spectral Data Fields

Element Description	Element Name	Data Type	Required	Document Section
Emitter Name	<code>EmitterName</code>	xs:string	Optional	4.5.17.1.1
Number of Wavelengths	<code>NumberWavelength</code>	xs:int	Yes	4.5.17.1.2
Multiplier	<code>Multiplier</code>	xs:decimal	Optional	4.5.17.1.3
Normalized	<code>Normalized</code>	xs:boolean	Optional	4.5.17.1.4
Timestamp	<code>Timestamp</code>	xs:int	Optional	4.5.17.1.5
Power Data	<code>PwrData</code>	XML Element	Yes	4.5.17.1.6
Quantum	<code>Quantum</code>	xs:boolean	Optional	4.5.17.1.7

4.5.17.1.1 Emitter Name Element The optional `EmitterName` element specifies the emitter name. For example, a multichannel theatrical luminaire may have separate color channels identified as “red,” “green,” “blue,” and “amber.”

4.5.17.1.2 Number of Wavelengths Element The required `NumberWavelength` element specifies the number of wavelengths for which data are reported.

4.5.17.1.3 Multiplier Element The optional `Multiplier` element, if present, indicates a floating-point multiplier that shall be applied to the spectral power measurements. A required attribute titled “comment” shall be used to describe the Multiplier. Examples of the comment attribute can be found in Annex C.

If the `Multiplier` element is not present, the default value is 1.0.

4.5.17.1.4 Normalized Element The optional `Normalized` element, if “true”, indicates that the spectral power data values have been normalized such that the maximum value is 1.0.

Note: If the `Normalized` element is not present, the default value is “false.”

4.5.17.1.5 Timestamp Element The optional `Timestamp` element is intended for luminous flux and color maintenance tests. The value represents the number of hours of operation from when the test procedure commenced.

4.5.17.1.6 Power Data Element The required `PwrData` element, when multiplied by the `Multiplier` element, specifies the emitter spectral data in watts per nanometer if the optional `Quantum` element is absent or "false". The element contains the value to be reported, and there is an attribute for the wavelength this value occurs at (in nanometers). For example:

```
<PwrData w="555.0">0.053</PwrData>
```

Note 1: Scientific notation (e.g., 1.23E-2) is allowed for the spectral power value.

Note 2: The combination of `Multiplier`, `Normalized`, and `Quantum` can have 8 possible cases:

1. `Normalized` = True, `Multiplier` Not Present, `Quantum` = False (or Not Present): `PwrData` elements are unitless and cannot be converted to W/nm in any way since `Multiplier` is missing, which defaults to 1.0.
2. `Normalized` = True, `Multiplier` Present, `Quantum` = False (or Not Present): `Multiplier` value is such that it will convert normalized data to W/nm.
3. `Normalized` = False (or Not Present), `Multiplier` Not Present, `Quantum` = False (or Not Present): `Multiplier` defaults to 1.0, and `PwrData` elements are already in W/nm.
4. `Normalized` = False (or Not Present), `Multiplier` Present, `Quantum` = False (or Not Present): `PwrData` elements when scaled by the `Multiplier` become W/nm, but are not scaled to a peak of 1.0.
5. `Normalized` = True, `Multiplier` Not Present, `Quantum` = True: `PwrData` elements are unitless and cannot be converted to $\mu\text{mol/s}\cdot\text{nm}$ in any way since `Multiplier` is missing, which defaults to 1.0.
6. `Normalized` = True, `Multiplier` Present, `Quantum` = True: `Multiplier` value is such that it will convert normalized data to $\mu\text{mol/s}\cdot\text{nm}$.
7. `Normalized` = False (or Not Present), `Multiplier` Not Present, `Quantum` = True: `Multiplier` defaults to 1.0, and `PwrData` elements are already in $\mu\text{mol/s}\cdot\text{nm}$.
8. `Normalized` = False (or Not Present), `Multiplier` Present, `Quantum` = True: `PwrData` elements when scaled by the `Multiplier` become $\mu\text{mol/s}\cdot\text{nm}$, but are not scaled to a peak of 1.0.

Note 3: If `Normalized` is "true" and `Multiplier` is not present, the spectral power data value is unitless.

Multiple `PwrData` elements are allowed.

4.5.17.1.7 Quantum Element The optional `Quantum` element, if "true", indicates that the spectral power data values, when multiplied by the `Multiplier` element, are reported in micromoles per second per nanometer ($\mu\text{mol/s}\cdot\text{nm}$) rather than watts per nanometer. The emitter spectral data then represents a spectral quantum distribution (SQD) rather than a spectral power distribution (SPD).

Note 1: A normalized SPD can be converted to a normalized SQD by multiplying each spectral power data value by $\lambda/\lambda_{\text{max}}$, where λ is the value wavelength and λ_{max} is the peak value wavelength, and renormalizing as required.

Note 2: If the `Quantum` element is not present, the default value is "false."

4.5.17.2 Angular Spectral Element The optional AngularSpectral element is the parent of angular spectral radiant intensity data. Elements are detailed in **Table 31**.

Table 31. Angular Spectral Intensity Data Fields

Element Description	Element Name	Data Type	Required	Document Section
Absolute	Absolute	xs:boolean	Optional	4.5.17.2.1
Symmetry	Symm	xs:string	Optional	4.5.17.2.2
Multiplier	Multiplier	xs:decimal	Optional	4.5.17.2.3
Number Measured	NumberMeasured	xs:int	Yes	4.5.17.2.4
Number of Horiz. Angles	NumberHorz	xs:int	Yes	4.5.17.2.5
Number of Vert. Angles	NumberVert	xs:int	Yes	4.5.17.2.6
Number of Wavelengths	NumberWavelength	xs:int	Yes	4.5.17.2.7
Intensity Data	IntData	XML Element	Yes	4.5.17.2.8

4.5.17.2.1 Absolute Element The optional Absolute element, if “true”, indicates that the angular spectral intensity measurements were performed using absolute photometry.

If the Absolute element is not present, the default value is “true”.

4.5.17.2.2 Symmetry Element The optional Symm element identifies the horizontal symmetry type. Values for CIE _ C and IES _ C gonioradiometer types only are shown in **Table 32**.

Table 32. Symmetry Types

Type	Description	Typical Horizontal Angle Range
Symm _ None	No symmetry about the vertical axis	0-360
Symm _ Bi _ 0	Bilateral symmetry about the vertical plane thru Horizontal 0-180 degree angles	0-180
Symm _ Bi _ 90	Bilateral symmetry about the vertical plane thru Horizontal 90-270 degree angles	90-270
Symm _ Quad	Quadrilateral symmetry about the vertical planes thru Horizontal 0-180 degree angles Horizontal 90-270 degree angles	0-90
Symm _ Full	Axial symmetry, identical vertical planes for every Horizontal angle	0
Symm _ Arbitrary	No symmetry, not a regular grid of horizontal and vertical angles	Undefined

If the Symm element is not present, the default value is Symm _ None.

Note 1: Symm _ Arbitrary indicates that the intensity data represent measurements at arbitrary horizontal and vertical angles, such as for a geodesic sphere, an example of which is shown in Annex D. The angles are identified in the IntData element using the h and v attributes for horizontal and vertical angle.

Note 2: If the Symm element is Symm_Arbitrary, the values of the NumberHorz and NumberVert elements shall be zero.

Note 3: Typical horizontal angles are the minimum range based on the indicated symmetry. The full data set can be included, but software reading these files may choose to present a warning indicating a potential mismatch of the indicated symmetry type and the IntData horizontal angles included in the file.

4.5.17.2.3 Multiplier Element The optional `Multiplier` element, if present, indicates a floating-point multiplier that shall be applied to the angular spectral intensity measurements. A required attribute titled “comment” shall be used to describe the Multiplier. Examples of the comment can be found in Annex C.

If the `Multiplier` element is not present, the default value is 1.0.

4.5.17.2.4 Number Measured Element The required `NumberMeasured` element specifies the number of angular spectral intensity measurements.

4.5.17.2.5 Number of Horiz. Angles Element The required `NumberHorz` element specifies the number of horizontal angles. The value shall be zero if the measurement coordinate system does not consist of a set of planes.

4.5.17.2.6 Number of Vert. Angles Element The required `NumberVert` element specifies the number of vertical angles. The value shall be zero if the measurement coordinate system does not consist of a set of planes.

4.5.17.2.7 Number of Wavelengths Element The required `NumberWavelength` element specifies the number of wavelengths.

4.5.17.2.8 Intensity Data Element The required `IntData` element, when multiplied by the `Multiplier` element, specifies the spectral radiant intensity data (in watts per steradian per nanometer) for the specified horizontal and vertical angles (in degrees). The element contains the value to be reported, and there are attributes for the horizontal and vertical angles and the wavelength (in nanometers) at which this value occurs. For example:

```
<IntData h="0.0" v="65.0" w="555.0">0.023</IntData>
```

Note 1: Scientific notation (e.g., 1.23E-2) is allowed for the angular spectral intensity value.

Note 2: The `Multiplier` element enables angular spectral intensity values to be reported in microwatts per steradian per nanometer or milliwatts per steradian per nanometer.

Multiple `IntData` elements are allowed.

4.5.18 Angular Color Element The optional `AngularColor` element is the parent of angular color data. Elements are detailed in **Table 33**.

Table 33. Angular Color Data Fields

Element Description	Element Name	Data Type	Required	Document Section
Absolute	<code>Absolute</code>	xs:boolean	Optional	4.5.18.1
Symmetry	<code>Symm</code>	xs:string	Optional	4.5.18.2
Multiplier	<code>Multiplier</code>	xs:decimal	Optional	4.5.18.3
Number Measured	<code>NumberMeasured</code>	xs:int	Yes	4.5.18.4
Number of Horiz. Angles	<code>NumberHorz</code>	xs:int	Yes	4.5.18.5
Number of Vert. Angles	<code>NumberVert</code>	xs:int	Yes	4.5.18.6
Color Data	<code>ColorData</code>	XML Element	Yes	4.5.18.7

4.5.18.1 Absolute Element The optional `Absolute` element, if “true”, indicates that the angular color measurements were performed using absolute photometry.

If the `Absolute` element is not present, the default value is “true”.

4.5.18.2 Symmetry Element The optional `Symm` element identifies the horizontal symmetry type. Valid values for CIE _ C and IES _ C gonioradiometer types only are shown in **Table 34**.

Table 34. Symmetry Types

Type	Description	Typical Horizontal Angle Range
<code>Symm _ None</code>	No symmetry about the vertical axis	0-360
<code>Symm _ Bi _ 0</code>	Bilateral symmetry about the vertical plane thru Horizontal 0-180 degree angles	0-180
<code>Symm _ Bi _ 90</code>	Bilateral symmetry about the vertical plane thru Horizontal 90-270 degree angles	90-270
<code>Symm _ Quad</code>	Quadrilateral symmetry about the vertical planes thru Horizontal 0-180 degree angles Horizontal 90-270 degree angles	0-90
<code>Symm _ Full</code>	Axial symmetry, identical vertical planes for every Horizontal angle	0
<code>Symm _ Arbitrary</code>	No symmetry, not a regular grid of horizontal and vertical angles	Undefined

If the `Symm` element is not present, the default value is `Symm _ None`.

Note 1: `Symm _ Arbitrary` indicates that the color data represent measurements at arbitrary horizontal and vertical angles, such as for a geodesic sphere, an example of which is shown in Annex D. The angles are identified in the `IntData` element using the `h` and `v` attributes for horizontal and vertical angle.

Note 2: If the `Symm` element is `Symm_Arbitrary`, the values of the `NumberHorz` and `NumberVert` elements shall be zero.

Note 3: Typical horizontal angles are the minimum range based on the indicated symmetry. The full data set can be included, but software reading these files may choose to present a warning indicating a potential mismatch of the indicated symmetry type and the `IntData` horizontal angles included in the file.

4.5.18.3 Multiplier Element The optional `Multiplier` element, if present, indicates a floating-point multiplier that shall be applied to the luminous intensity (CIE Y) measurements. A required attribute titled “comment” shall be used to describe the `Multiplier`. Examples of the `comment` attribute can be found in Annex C.

If the `Multiplier` element is not present, the default value is 1.0.

4.5.18.4 Number Measured Element The required `NumberMeasured` element specifies the number of angular color measurements.

4.5.18.5 Number of Horiz. Angles Element The `NumberHorz` element specifies the number of horizontal angles. The value shall be zero if the measurement coordinate system does not consist of a set of planes.

4.5.18.6 Number of Vert. Angles Element The `NumberVert` element specifies the number of vertical angles. The value shall be zero if the measurement coordinate system does not consist of a set of planes.

4.5.18.7 Color Data Element The required `ColorData` element specifies the CIE x,y chromaticity coordinates and, when multiplied by the `Multiplier` element, the luminous intensity data (CIE Y, stated in candelas) for the specified horizontal and vertical angles (in degrees). The element contains the value to be reported, and there are attributes for the horizontal and vertical angles, and CIE x,y chromaticity coordinates at which the value occurs. For example:

```
<ColorData h="0.0" v="65.0" x="0.435" y="0.401">65.0</ColorData>
```

Note 1: Scientific notation (e.g., 1.23E-2) is allowed for the angular luminous intensity value.

Note 2: The `Multiplier` element enables luminous intensity values to be reported in candela per kilolumen.

Multiple `ColorData` elements are allowed.

4.5.19 Illuminance Data Element The optional `IllumData` element is the parent of illuminance data. This element contains information that is specific to illuminance data. Elements are detailed in **Table 35**.

Table 35. Illuminance Data Fields

Element Description	Element Name	Data Type	Required	Document Section
Absolute	<code>Absolute</code>	xs:boolean	Optional	4.5.19.1
Multiplier	<code>Multiplier</code>	xs:decimal	Optional	4.5.19.2
Number of Planes	<code>NumberPlanes</code>	xs:int	Yes	4.5.19.3
Plane Data	<code>PlaneData</code>	XML Element	Yes	4.5.19.4

4.5.19.1 Absolute Element The optional `Absolute` element if “true”, indicates that the illuminance measurements were performed using absolute photometry.

If the `Absolute` element is not present, the default value is “true”.

4.5.19.2 Multiplier Element The optional `Multiplier` element, if present, indicates a floating-point multiplier that shall be applied to the illuminance measurements. An required attribute titled “comment” shall be used to describe the Multiplier. Examples of the comment attribute can be found in Annex C.

If the `Multiplier` element is not present, the default value is 1.0.

4.5.19.3 Number of Planes Element The required `NumberPlanes` element specifies the number of measurement planes.

4.5.19.4 Plane Data Element The required `PlaneData` element specifies the illuminance data for the specified measurement plane. Elements are detailed in **Table 36**.

Multiple `PlaneData` elements are allowed.

Table 36. Plane Data Fields

Element Description	Element Name	Data Type	Required	Document Section
Plane Normal	<code>PlaneNormal</code>	XML Element	Optional	4.5.19.4.1
Number Measured	<code>NumberMeasured</code>	xs:int	Yes	4.5.19.4.2
Illuminance	<code>Illum</code>	XML Element	Yes	4.5.19.4.3

4.5.19.4.1 Plane Normal Element The optional `PlaneNormal` element specifies coordinates of the unit plane normal with respect to the luminaire dimensions coordinate system. For example:

```
<PlaneNormal x="-1.0" y="0.0 z="0.0"/>
```

Note: The plane can be oriented arbitrarily with respect to the emitter.

If absent, the unit plane normal coordinates are assumed to be aligned with zenith:

```
<PlaneNormal x="0.0" y="0.0 z="1.0"/>
```

4.5.19.4.2 Number Measured Element The required `NumberMeasured` element specifies the number of illuminance measurements for the specified plane.

4.5.19.4.3 Illuminance Element The required `Illum` element, when multiplied by the `Multiplier` element, specifies illuminance data (in candelas per square meter) for the specified position. The element contains the value to be reported, and there are attributes for the plane position (in meters) at which this value occurs. For example:

```
<Illum x="0.0" y="1.5" z="1.2">55</Illum>
```

Note 1: Scientific notation (e.g., 1.23E-2) is allowed for the illuminance value.

Note 2: The `Multiplier` element enables illuminance values to be reported in candelas per square meter per kilolumen.

Multiple `Illum` elements are allowed.

4.5.20 Irradiance Data Element The optional `IrradData` element is the parent of irradiance data. This element contains information that is specific to the irradiance data. Elements are detailed in **Table 37**.

Table 37. Irradiance Data Fields

Element Description	Element Name	Data Type	Required	Document Section
Absolute	<code>Absolute</code>	xs:boolean	Optional	4.5.20.1
Multiplier	<code>Multiplier</code>	xs:decimal	Optional	4.5.20.2
Minimum Wavelength	<code>MinWavelength</code>	xs:decimal	Yes	4.5.20.3
Maximum Wavelength	<code>MaxWavelength</code>	xs:decimal	Yes	4.5.20.4
Number of Planes	<code>NumberPlanes</code>	xs:int	Yes	4.5.20.5
Plane Data	<code>PlaneData</code>	XML Element	Yes	4.5.20.6

4.5.20.1 Absolute Element The optional `Absolute` element, if “true”, indicates that the irradiance measurements were performed using absolute photometry.

If the `Absolute` element is not present, the default value is “true”.

4.5.20.2 Multiplier Element The optional `Multiplier` element, if present, indicates a floating-point multiplier that shall be applied to the irradiance measurements. An required attribute titled “comment” shall be used to describe the `Multiplier`. Examples of the comment attribute can be found in Annex C.

If the `Multiplier` element is not present, the default value is 1.0.

4.5.20.3 Minimum Wavelength Element The required `MinWavelength` element specifies the minimum wavelength of the irradiance measurements, in nanometers.

4.5.20.4 Maximum Wavelength Element The required `MaxWavelength` element specifies the maximum wavelength of the irradiance measurements, in nanometers.

4.5.20.5 Number of Planes Element The required `NumberPlanes` element specifies the number of measurement planes.

4.5.20.6 Plane Data Element The required `PlaneData` element specifies the irradiance data for the specified measurement plane. Elements are detailed in **Table 38**.

Multiple `PlaneData` elements are allowed.

Table 38. Plane Data Fields

Element Description	Element Name	Data Type	Required	Document Section
Plane Normal	<code>PlaneNormal</code>	XML Element	Optional	4.5.20.6.1
Number Measured	<code>NumberMeasured</code>	<code>xs:int</code>	Yes	4.5.20.6.2
Irradiance	<code>Irrad</code>	XML Element	Yes	4.5.20.6.3

4.5.20.6.1 Plane Normal Element The optional `PlaneNormal` element specifies coordinates of the unit plane normal with respect to the luminaire dimensions coordinate system. For example:

```
<PlaneNormal x="-1.0" y="0.0 z="0.0"/>
```

Note: The plane can be oriented arbitrarily with respect to the emitter.

If absent, the unit plane normal coordinates are assumed to be aligned with zenith:

```
<PlaneNormal x="0.0" y="0.0 z="1.0"/>
```

4.5.20.6.2 Number Measured Element The required `NumberMeasured` element specifies the number of irradiance measurements for the specified plane.

4.5.20.6.3 Irradiance Element The required `Irrad` element, when multiplied by the `Multiplier` element, specifies irradiance data (in watts per square meter) for the specified position. The element contains the value to be reported, and there are attributes for the plane position (in meters) at which this value occurs. For example:

```
<Irrad x="0.0" y="1.5" z="1.2">55</Irrad>
```

Note 1: Scientific notation (e.g., 1.23E-2) is allowed for the irradiance value.

Note 2: The `Multiplier` element enables irradiance values to be reported in microwatts per square meter or milliwatts per square meter.

Multiple `Irrad` elements are allowed.

4.5.21 Photon Flux Density Data Element

The optional `PFDDData` element is the parent of photon flux density data. This element contains information that is specific to the photon flux density data. Elements are detailed in **Table 39**.

Table 39. Photon Flux Density Data Fields

Element Description	Element Name	Data Type	Required	Document Section
Absolute	<code>Absolute</code>	<code>xs:boolean</code>	Optional	4.5.21.1
Multiplier	<code>Multiplier</code>	<code>xs:decimal</code>	Optional	4.5.21.2
Minimum Wavelength	<code>MinWavelength</code>	<code>xs:decimal</code>	Yes	4.5.21.3
Maximum Wavelength	<code>MaxWavelength</code>	<code>xs:decimal</code>	Yes	4.5.21.4
Number of Planes	<code>NumberPlanes</code>	<code>xs:int</code>	Yes	4.5.21.5
Plane Data	<code>PlaneData</code>	XML Element	Yes	4.5.21.6

4.5.21.1 Absolute Element The optional `Absolute` element, if “true”, indicates that the irradiance measurements were performed using absolute photometry.

If the `Absolute` element is not present, the default value is “true”.

4.5.21.2 Multiplier Element The optional `Multiplier` element, if present, indicates a floating-point multiplier that shall be applied to the photon flux density measurements. An required attribute titled “comment” shall be used to describe the Multiplier. Examples of the comment attribute can be found in Annex C.

If the `Multiplier` element is not present, the default value is 1.0.

4.5.21.3 Minimum Wavelength Element The required `MinWavelength` element specifies the minimum wavelength of the photon flux density measurements, in nanometers.

4.5.21.4 Maximum Wavelength Element The required `MaxWavelength` element specifies the maximum wavelength of the photon flux density measurements, in nanometers.

4.5.21.5 Number of Planes Element The required `NumberPlanes` element specifies the number of measurement planes.

4.5.21.6 Plane Data Element The required `PlaneData` element specifies the photon flux density data for the specified measurement plane. Elements are detailed in **Table 40**.

Multiple `PlaneData` elements are allowed.

Table 40. Plane Data Fields

Element Description	Element Name	Data Type	Required	Document Section
Plane Normal	<code>PlaneNormal</code>	XML Element	Optional	4.5.21.6.1
Number Measured	<code>NumberMeasured</code>	xs:int	Yes	4.5.21.6.2
Photon Flux Density	<code>PFD</code>	XML Element	Yes	4.5.21.6.3

4.5.21.6.1 Plane Normal Element The optional `PlaneNormal` element specifies coordinates of the unit plane normal with respect to the luminaire dimensions coordinate system. For example:

```
<PlaneNormal x="-1.0" y="0.0" z="0.0"/>
```

Note: The plane can be oriented arbitrarily with respect to the emitter.

If absent, the unit plane normal coordinates are assumed to be aligned with zenith:

```
<PlaneNormal x="0.0" y="0.0" z="1.0"/>
```

4.5.21.6.2 Number Measured Element The required `NumberMeasured` element specifies the number of photon flux density measurements for the specified plane.

4.5.21.6.3 Photon Flux Density Element The required `PFD` element, when multiplied by the `Multiplier` element, specifies photon flux density data (in micromoles per second per square meter) for the specified position. The element contains the value to be reported, and there are attributes for the plane position (in meters) at which this value occurs. For example:

```
<PFD x="0.0" y="1.5" z="1.2">55</PFD>
```

Note: Scientific notation (e.g., 1.23E-2) is allowed for the photon flux density value.

Multiple PFD elements are allowed.

4.5.22 Spectral Irradiance Data Element The optional `SpecIrradData` element is the parent of spectral irradiance data. This element contains information that is specific to the spectral irradiance data. Elements are detailed in **Table 41**.

Table 41. Spectral Irradiance Data Fields

Element Description	Element Name	Data Type	Required	Document Section
Absolute	<code>Absolute</code>	xs:boolean	Optional	4.5.22.1
Multiplier	<code>Multiplier</code>	xs:decimal	Optional	4.5.22.2
Number of Planes	<code>NumberPlanes</code>	xs:int	Yes	4.5.22.3
Plane Data	<code>PlaneData</code>	XML Element	Yes	4.5.22.4

4.5.22.1 Absolute Element The optional `Absolute` element, if “true”, indicates that the spectral irradiance measurements were performed using absolute photometry.

If the `Absolute` element is not present, the default value is “true”.

4.5.22.2 Multiplier Element The optional `Multiplier` element, if present indicates a floating-point multiplier that shall be applied to the spectral irradiance measurements. An required attribute titled “comment” shall be used to describe the Multiplier. Examples of the comment attribute can be found in Annex C.

If the `Multiplier` element is not present, the default value is 1.0.

4.5.22.3 Number of Planes Element The required `NumberPlanes` element specifies the number of measurement planes.

4.5.22.4 Plane Data Element

The required `PlaneData` element specifies the spectral irradiance data for the specified measurement plane. Elements are detailed in **Table 42**.

Multiple `PlaneData` elements are allowed.

Table 42. Plane Data Fields

Element Description	Element Name	Data Type	Required	Document Section
Plane Normal	<code>PlaneNormal</code>	XML Element	Optional	4.5.22.4.1
Number Measured	<code>NumberMeasured</code>	xs:int	Yes	4.5.22.4.2
Number of Wavelengths	<code>NumberWavelength</code>	xs:int	Yes	4.5.22.4.3
Spectral Irradiance	<code>SIrrad</code>	XML Element	Yes	4.5.22.4.4

4.5.22.4.1 Plane Normal Element The optional `PlaneNormal` element specifies coordinates of the unit plane normal with respect to the luminaire dimensions coordinate system. For example:

```
<PlaneNormal x="-1.0" y="0.0 z="0.0"/>
```

Note: The plane can be oriented arbitrarily with respect to the emitter.

If absent, the unit plane normal coordinates are assumed to be aligned with zenith:

```
<PlaneNormal x="0.0" y="0.0 z="1.0"/>
```

4.5.22.4.2 Number Measured Element The required `NumberMeasured` element specifies the number of spectral irradiance measurements for the specified plane.

4.5.22.4.3 Number of Wavelengths Element The required `NumberWavelength` element specifies the number of wavelengths.

4.5.22.4.4 Spectral Irradiance Element The required `SIrrad` element, when multiplied by the `Multiplier` element, specifies spectral irradiance data (in watts per square meter per nanometer) for the specified position. The element contains the value to be reported, and there are attributes for the plane position (in meters) and wavelength (in nanometers) at which this value occurs. For example:

```
<SIrrad x="0.0" y="1.5" z="1.2" w="555.0">55</SIrrad >
```

Note 1: Scientific notation (e.g., 1.23E-2) is allowed for the spectral irradiance value.

Note 2: The `Multiplier` element enables spectral irradiance values to be reported in microwatts per square meter per nanometer or milliwatts per square meter per nanometer.

Multiple `SIrrad` elements are allowed.

4.5.23 Channels Element A luminaire may have multiple emitters with different spectral power distributions, such as a theatrical luminaire with red, green, blue, and amber light emitting diodes. If the intensity of each emitter can be independently controlled (that is, dimmed) without significantly changing the spatial intensity distribution, a “channel multiplier” can be applied to the measured or simulated intensity distribution to represent the intensity distribution when the channel is tested at a specific correlated color temperature.

The optional `Channels` element is the parent of emitter channels data. Elements are detailed in **Table 43**.

Table 43. Channels Fields

Element Description	Element Name	Data Type	Required	Document Section
Number of Channels	<code>NumberChannels</code>	xs:int	Yes	4.5.23.1
Channel Multiplier	<code>ChannelMult</code>	XML Element	Yes	4.5.23.2

4.5.23.1 Number of Channels Element The required `NumberChannels` element specifies the number of independent emitter channels.

4.5.23.2 Channel Multiplier Element The required `ChannelMult` element specifies an independent emitter channel multiplier that is applied to the emitter intensity values or flux value. For example:

```
<ChannelMult name="red">0.63</ChannelMult> ,
```

where the channel names should be the same as emitter spectral names (see **Section 4.5.17.1.1**) if multiple `EmitterSpectral` elements are present.

Multiple `ChannelMult` elements are allowed.

4.5.24 Emission Areas Element The optional `EmissionAreas` element is the parent of emission areas data. Elements are detailed in **Table 44**.

Table 44. Emission Face Fields

Element Description	Element Name	Data Type	Required	Document Section
Top Face	TopFace	XML Element	Optional	4.5.24.1
Bottom Face	BottomFace	XML Element	Optional	4.5.24.2
C0 Face	C0Face	XML Element	Optional	4.5.24.3
C90 Face	C90Face	XML Element	Optional	4.5.24.4
C180 Face	C180Face	XML Element	Optional	4.5.24.5
C270 Face	C270Face	XML Element	Optional	4.5.24.6

Note: For cylindrical shapes, the planar emission faces are orthographically projected onto the cylindrical face.

4.5.24.1 Top Face Element The optional `TopFace` element is the parent of top face emission areas data. Elements are detailed in **Table 45**.

4.5.24.1.1 Number Top Element The required `NumberTop` element specifies the number of top face emission areas.

Table 45. Top Face Fields

Element Description	Element Name	Data Type	Required	Document Section
Number Top	NumberTop	xs:int	Yes	4.5.24.1.1
Top Area	TopArea	XML Element	Yes	4.5.24.1.2

4.5.24.1.2 Top Area Element The required `TopArea` element is the parent of top face emission area data. Elements are detailed in **Table 46**.

Multiple `TopArea` elements are allowed.

Table 46. Top Face Emission Area Fields

Element Description	Element Name	Data Type	Required	Document Section
Length	Length	xs:decimal	Yes	4.5.24.1.2.1
Width	Width	xs:decimal	Yes	4.5.24.1.2.2
Length Offset	LengthOffset	xs:decimal	Yes	4.5.24.1.2.3
Width Offset	WidthOffset	xs:decimal	Yes	4.5.24.1.2.4
Circular	Circular	xs:boolean	Optional	4.5.24.1.2.5

4.5.24.1.2.1 Length Element The required `Length` element specifies the length of the top face emission area, in meters.

4.5.24.1.2.2 Width Element The required `Width` element specifies the width of the top face emission area, in meters.

4.5.24.1.2.3 Length Offset Element The required `LengthOffset` element specifies the offset of the top face emission area from the geometric center of the luminaire along the length (y) axis, in meters.

4.5.24.1.2.4 Width Offset Element The required `WidthOffset` element specifies the offset of the top face emission area from the geometric center of the luminaire along the width (x) axis, in meters.

4.5.24.1.2.5 Circular Element The optional Boolean `Circular` element, if “true”, indicates that the top face emission area is circular rather than rectangular. (If the length and width element values are not equal, the area is considered to be elliptical.)

4.5.24.2 Bottom Face Element The optional `BottomFace` element is the parent of bottom face emission areas data. Elements are detailed in **Table 47**.

Table 47. Bottom Face Fields

Element Description	Element Name	Data Type	Required	Document Section
Number Bottom	NumberBottom	xs:int	Yes	4.5.24.2.1
Bottom Area	BottomArea	XML Element	Yes	4.5.24.2.2

4.5.24.2.1 Number Bottom Element The required `NumberBottom` element specifies the number of bottom face emission areas.

4.5.24.2.2 Bottom Area Element The required `BottomArea` element is the parent of the bottom face emission area data. Elements are detailed in **Table 48**.

Multiple `BottomArea` elements are allowed.

Table 48. Bottom Face Emission Area Fields

Element Description	Element Name	Data Type	Required	Document Section
Length	Length	xs:decimal	Yes	4.5.24.2.2.1
Width	Width	xs:decimal	Yes	4.5.24.2.2.2
Length Offset	LengthOffset	xs:decimal	Yes	4.5.24.2.2.3
Width Offset	WidthOffset	xs:decimal	Yes	4.5.24.2.2.4
Circular	Circular	xs:boolean	Optional	4.5.24.2.2.5

4.5.24.2.2.1 Length Element The required `Length` element specifies the length of the bottom face emission area, in meters.

4.5.24.2.2.2 Width Element The required `Width` element specifies the width of the bottom face emission area, in meters.

4.5.24.2.2.3 Length Offset Element The required `LengthOffset` element specifies the offset of the bottom face emission area from the geometric center of the luminaire along the length (y) axis, in meters.

4.5.24.2.2.4 Width Offset Element The required `WidthOffset` element specifies the offset of the bottom face emission area from the geometric center of the luminaire along the width (x) axis, in meters.

4.5.24.2.2.5 Circular Element The optional Boolean `Circular` element, if “true”, indicates that the bottom face emission area is circular rather than rectangular. (If the length and width element values are not equal, the area is considered to be elliptical.)

4.5.24.3 C0 Face Element The optional `C0Face` element is the parent of C0 face emission areas data. Elements are detailed in **Table 49**.

Table 49. C0 Face Fields

Element Description	Element Name	Data Type	Required	Document Section
Number C0	NumberC0	xs:int	Yes	4.5.24.3.1
C0 Area	C0Area	XML Element	Yes	4.5.24.3.2

4.5.24.3.1 Number C0 Element The required `NumberC0` element specifies the number of C0 face emission areas.

4.5.24.3.2 C0 Area Element The required `C0Area` element is the parent of C0 face emission area data. Elements are detailed in **Table 50**.

Multiple `C0Area` elements are allowed.

4.5.24.3.2.1 Length Element The required `Length` element specifies the length of the C0 face emission area, in meters.

Table 50. C0 Face Emission Area Fields

Element Description	Element Name	Data Type	Required	Document Section
Length	Length	xs:decimal	Yes	4.5.24.3.2.1
Height	Height	xs:decimal	Yes	4.5.24.3.2.2
Length Offset	LengthOffset	xs:decimal	Yes	4.5.24.3.2.3
Height Offset	HeightOffset	xs:decimal	Yes	4.5.24.3.2.4
Circular	Circular	xs:boolean	Optional	4.5.24.3.2.5

4.5.24.3.2.2 Height Element The required `Height` element specifies the height of the C0 face emission area, in meters.

4.5.24.3.2.3 Length Offset Element The required `LengthOffset` element specifies the offset of the C0 face emission area from the geometric center of the luminaire along the length (y) axis, in meters.

4.5.24.3.2.4 Height Offset Element The required `HeightOffset` element specifies the offset of the C0 face emission area from the geometric center of the luminaire along the height (z) axis, in meters.

4.5.24.3.2.5 Circular Element The optional Boolean `Circular` element, if “true”, indicates that the C0 face emission area is circular rather than rectangular. (If the length and width element values are not equal, the area is considered to be elliptical.)

4.5.24.4 C90 Face Element The optional `C90Face` element is the parent of C90 face emission areas data. Elements are detailed in **Table 51**.

Table 51. C90Face Fields

Element Description	Element Name	Data Type	Required	Document Section
Number C90	NumberC90	xs:int	Yes	4.5.24.4.1
C90 Area	C90Area	XML Element	Yes	4.5.24.4.2

4.5.24.4.1 Number C90 Element The required `NumberC90` element specifies the number of C90 face emission areas.

4.5.24.4.2 C90 Area Element The required `C90Area` element is the parent of C90 face emission area data. Elements are detailed in **Table 52**.

Multiple `C90Area` elements are allowed.

Table 52. C90 Face Emission Area Fields

Element Description	Element Name	Data Type	Required	Document Section
Width	Width	xs:decimal	Yes	4.5.24.4.2.1
Height	Height	xs:decimal	Yes	4.5.24.4.2.3
Width Offset	WidthOffset	xs:decimal	Yes	4.5.24.4.2.3
Height Offset	HeightOffset	xs:decimal	Yes	4.5.24.4.2.4
Circular	Circular	xs:boolean	Optional	4.5.24.4.2.5

4.5.24.4.2.1 Width Element The required `Width` element specifies the width of the C90 face emission area, in meters.

4.5.24.4.2.2 Height Element The required `Height` element specifies the height of the C90 face emission area, in meters.

4.5.24.4.2.3 Width Offset Element The required `WidthOffset` element specifies the offset of the C90 face emission area from the geometric center of the luminaire along the width (x) axis, in meters.

4.5.24.4.2.4 Height Offset Element The required `HeightOffset` element specifies the offset of the C90 face emission area from the geometric center of the luminaire along the height (z) axis, in meters.

4.5.24.4.2.5 Circular Element The optional Boolean `Circular` element, if “true”, indicates that the C90 face emission area is circular rather than rectangular. (If the length and width element values are not equal, the area is considered to be elliptical.)

4.5.24.5 C180 Face Element The optional `C180Face` element is the parent of C180 face emission areas data. Elements are detailed in **Table 53**.

Table 53. C180 Face Fields

Element Description	Element Name	Data Type	Required	Document Section
Number C180	NumberC180	xs:int	Yes	4.5.24.5.1
C180 Area	C180Area	XML Element	Yes	4.5.24.5.2

4.5.24.5.1 Number C180 Element The required `NumberC180` element specifies the number of C180 face emission areas.

4.5.24.5.2 C180 Area Element The required `C180Area` element is the parent of C180 face emission area data. Elements are detailed in **Table 54**.

Multiple `C180Area` elements are allowed.

Table 54. C180 Face Emission Area Fields

Element Description	Element Name	Data Type	Required	Document Section
Length	Length	xs:decimal	Yes	4.5.24.5.2.1
Height	Height	xs:decimal	Yes	4.5.24.5.2.2
Length Offset	LengthOffset	xs:decimal	Yes	4.5.24.5.2.3
Height Offset	HeightOffset	xs:decimal	Yes	4.5.24.5.2.4
Circular	Circular	xs:boolean	Optional	4.5.24.5.2.5

4.5.24.5.2.1 Length Element The required `Length` element specifies the length of the C180 face emission area, in meters.

4.5.24.5.2.2 Height Element The required `Height` element specifies the height of the C180 face emission area, in meters.

4.5.24.5.2.3 Length Offset Element The required `LengthOffset` element specifies the offset of the C180 face emission area from the geometric center of the luminaire along the length (y) axis, in meters.

4.5.24.5.2.4 Height Offset Element The required `HeightOffset` element specifies the offset of the C180 face emission area from the geometric center of the luminaire along the height (z) axis, in meters.

4.5.24.5.2.5 Circular Element The optional Boolean `Circular` element, if “true”, indicates that the C180 face emission area is circular rather than rectangular. (If the length and width element values are not equal, the area is considered to be elliptical.)

4.5.24.6 C270 Face Element The optional `C270Face` element is the parent of C270 face emission areas data. Elements are detailed in **Table 55**.

Table 55. C270 Face Fields

Element Description	Element Name	Data Type	Required	Document Section
Number C270	NumberC270	xs:int	Yes	4.5.24.6.1
C270 Area	C270Area	XML Element	Yes	4.5.24.6.2

4.5.24.6.1 Number C270 Element The required `NumberC270` element specifies the number of C270 face emission areas.

4.5.24.6.2 C270 Area Element The required `C270Area` element is the parent of C270 face emission area data. Elements are detailed in **Table 56**.

Multiple `C270Area` elements are allowed.

Table 56. C270 Face Emission Area Fields

Element Description	Element Name	Data Type	Required	Document Section
Width	Width	xs:decimal	Yes	4.5.24.6.2.1
Height	Height	xs:decimal	Yes	4.5.24.6.2.2
Width Offset	WidthOffset	xs:decimal	Yes	4.5.24.6.2.3
Height Offset	HeightOffset	xs:decimal	Yes	4.5.24.6.2.4
Circular	Circular	xs:boolean	Optional	4.5.24.6.2.5

4.5.24.6.2.1 Width Element The required `Width` element specifies the width of the C270 face emission area, in meters.

4.5.24.6.2.2 Height Element The required `Height` element specifies the height of the C270 face emission area, in meters.

4.5.24.6.2.3 Width Offset Element The required `WidthOffset` element specifies the offset of the C270 face emission area from the geometric center of the luminaire along the width (x) axis, in meters.

4.5.24.6.2.4 Height Offset Element The required `HeightOffset` element specifies the offset of the C270 face emission area from the geometric center of the luminaire along the height (z) axis, in meters.

4.5.24.6.2.5 Circular Element The optional Boolean `Circular` element, if “true”, indicates that the C270 face emission area is circular rather than rectangular. (If the length and width element values are not equal, the area is considered to be elliptical.)

4.5.25 Emitter Center Element The optional `EmitterCenter` element is the parent of emitter center data. Elements are detailed in **Table 57**.

Table 57. Emitter Center Fields

Element Description	Element Name	Data Type	Required	Document Section
Length Offset	<code>LengthOffset</code>	xs:decimal	Yes	4.5.25.1
Width Offset	<code>WidthOffset</code>	xs:decimal	Yes	4.5.25.2
Height Offset	<code>HeightOffset</code>	xs:decimal	Yes	4.5.25.3

If the `EmitterCenter` element is absent, the emitter center is assumed to be coincident with the geometric center.

4.5.25.1 Length Offset Element The required `LengthOffset` element specifies the offset of the emitter center from the geometric center of the luminaire along the length (y) axis, in meters.

4.5.25.2 Width Offset Element The required `WidthOffset` element specifies the offset of the emitter center from the geometric center of the luminaire along the width (x) axis, in meters.

4.5.25.3 Height Offset Element The required `HeightOffset` element specifies the offset of the emitter center from the geometric center of the luminaire along the height (z) axis, in meters.

4.5.26 Regulatory Element The optional `Regulatory` element is the parent of regulatory data. Elements are detailed in **Table 58**.

Table 58. Regulatory Fields

Element Description	Element Name	Data Type	Required	Document Section
Input Wattage	<code>InputWattage</code>	xs:string	Optional	4.5.5
Power Factor	<code>PowerFactor</code>	xs:string	Optional	4.5.6
Ballast Factor	<code>BallastFactor</code>	xs:string	Optional	4.5.7
Correlated Color Temperature (CCT)	<code>ColorTemperature</code>	xs:string	Optional	4.5.9
CIE CRI	<code>CIE_CRI</code>	xs:string	Optional	4.5.10.1
IES TM-30	<code>IES_TM30</code>	xs:string	Optional	4.5.10.2
Duv	<code>Duv</code>	xs:string	Optional	4.5.11
S/P Ratio	<code>SPRatio</code>	xs:string	Optional	4.5.12
Luminous Intensity	<code>LuminousIntensity</code>	xs:string	Optional	4.5.14.1.6
Luminous Flux	<code>LuminousFlux</code>	xs:string	Optional	4.5.14.2
Radiant Intensity	<code>RadiantIntensity</code>	xs:string	Optional	4.5.15.3.5
Radiant Flux	<code>RadiantFlux</code>	xs:string	Optional	4.5.15.4
Photon Intensity	<code>PhotonIntensity</code>	xs:string	Optional	4.5.16.3.5
Photon Flux	<code>PhotonFlux</code>	xs:string	Optional	4.5.16.4

(continued on next page)

Table 58. Regulatory Fields *(continued)*

Element Description	Element Name	Data Type	Required	Document Section
Spectral Power	SpectralPower	xs:string	Optional	4.5.17.1.5
Spectral Intensity	SpectralIntensity	xs:string	Optional	4.5.17.2.7
Angular Color	AngularColor	xs:string	Optional	4.5.18.5
Illuminance	Illuminance	xs:string	Optional	4.5.19.2.4
Irradiance	Irradiance	xs:string	Optional	4.5.20.4.4
Photon Flux Density	PhotonFluxDensity	xs:string	Optional	4.5.21.4.4
Spectral Irradiance	SpectralIrradiance	xs:string	Optional	4.5.22.2.5

Each optional `Regulatory` element shall be identified as having measured, nominal, or rated values, as listed in **Table 59**.

Table 59. Regulatory Value Types

Type	Description
Measured	Measured value(s)
Nominal	Nominal value(s)
Rated	Rated value(s)

A *rated* value is the value of a quantity used for specification purposes, established for a specified set of operating conditions of the product.

A nominal value is an approximate quantity value used to designate or identify a product.

If a `Regulatory` element is absent, its values are assumed to be measured or calculated from measured data.

4.6 Custom Data Element

The optional `CustomData` element is the parent of custom data. Elements are detailed in **Table 60**.

The intent of the `CustomData` element is to provide a means of including custom data within the luminaire optical-data document. For example, a company or government agency may require additional information that cannot be represented within the base XML schema (see **Section 4.7**).

Multiple `CustomData` elements are allowed.

Table 60. Custom Data Fields

Element Description	Element Name	Data Type	Required	Document Section
Name	Name	xs:string	Yes	4.6.1
Unique Identifier	UniqueIdentifier	xs:string	Yes	4.6.2
Any Data	[Not applicable]	XML Element	Optional	4.6.3

4.6.1 Name Element The required `Name` element specifies the name describing the custom data, for example “Italian CAM”.

4.6.2 Unique Identifier Element The required `UniqueIdentifier` element contains a Universally Unique Identifier (UUID), as defined by RFC 4122 (see **Section 2.15**). Most scientific programming language libraries include functions that will automatically generate UUIDs.

A UUID is a unique 128-bit value expressed as a string with 32 hexadecimal digits in the format xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx. For example:

E1EC2128-3AEB-1069-A24E-98712B54306F

The purpose of this UUID is to prevent name collisions between different instances of the `Name` element.

4.6.3 Any Data Element The base XML schema (see **Section 4.7**) allows an arbitrary sequence of XML elements within the `CustomData` element using an `<any>` declaration with a `##any` namespace attribute to instruct a validating parser to ignore the element sequence for validation purposes.

4.7 XML Schema

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">

  <!-- Common attribute definitions -->
  <xs:attribute name="angle" type="xs:decimal"/>
  <xs:attribute name="h" type="xs:decimal"/>
  <xs:attribute name="v" type="xs:decimal"/>
  <xs:attribute name="w" type="xs:decimal"/>
  <xs:attribute name="x" type="xs:decimal"/>
  <xs:attribute name="y" type="xs:decimal"/>
  <xs:attribute name="z" type="xs:decimal"/>

  <!-- Common simple element definitions -->
  <xs:element name="Circular" type="xs:boolean"/>
  <xs:element name="Height" type="xs:decimal"/>
  <xs:element name="HeightOffset" type="xs:decimal"/>
  <xs:element name="Length" type="xs:decimal"/>
  <xs:element name="LengthOffset" type="xs:decimal"/>
  <xs:element name="MaxWavelength" type="xs:decimal"/>
  <xs:element name="MeasurementEquipment" type="xs:string"/>
  <xs:element name="MinWavelength" type="xs:decimal"/>
  <xs:element name="Multiplier" type="xs:float"/>
  <xs:element name="NumberHorz" type="xs:int"/>
  <xs:element name="NumberMeasured" type="xs:int"/>
  <xs:element name="NumberPlanes" type="xs:int"/>
  <xs:element name="NumberVert" type="xs:int"/>
  <xs:element name="NumberWavelength" type="xs:int"/>
  <xs:element name="Width" type="xs:decimal"/>
  <xs:element name="WidthOffset" type="xs:decimal"/>

  <!-- Common element definitions -->
  <xs:complexType name="IntDataType2">
    <xs:simpleContent>
      <xs:extension base="xs:float">
        <xs:attribute ref="h"/>
        <xs:attribute ref="v"/>
      </xs:extension>
    </xs:simpleContent>
  </xs:complexType>
  <xs:complexType name="IntDataType3">
    <xs:simpleContent>
      <xs:extension base="xs:float">
```

```

        <xs:attribute ref="h"/>
        <xs:attribute ref="v"/>
        <xs:attribute ref="w"/>
    </xs:extension>
</xs:simpleContent>
</xs:complexType>
<xs:complexType name="IntDataType4">
    <xs:simpleContent>
        <xs:extension base="xs:float">
            <xs:attribute ref="x"/>
            <xs:attribute ref="y"/>
        </xs:extension>
    </xs:simpleContent>
</xs:complexType>
<xs:complexType name="PlaneNormalType">
    <xs:simpleContent>
        <xs:extension base="xs:decimal">
            <xs:attribute ref="x"/>
            <xs:attribute ref="y"/>
            <xs:attribute ref="z"/>
        </xs:extension>
    </xs:simpleContent>
</xs:complexType>
<xs:complexType name="PwrDataType">
    <xs:simpleContent>
        <xs:extension base="xs:float">
            <xs:attribute ref="w"/>
        </xs:extension>
    </xs:simpleContent>
</xs:complexType>
<xs:simpleType name="RegulatoryValue">
    <xs:restriction base="xs:string">
        <xs:enumeration value="Measured"/>
        <xs:enumeration value="Nominal"/>
        <xs:enumeration value="Rated"/>
    </xs:restriction>
</xs:simpleType>
<xs:simpleType name="SymmType">
    <xs:restriction base="xs:string">
        <xs:enumeration value="Symm _ None"/>
        <xs:enumeration value="Symm _ Bi _ 0"/>
        <xs:enumeration value="Symm _ Bi _ 90"/>
        <xs:enumeration value="Symm _ Quad"/>
        <xs:enumeration value="Symm _ Full"/>
        <xs:enumeration value="Symm _ Arbitrary"/>
    </xs:restriction>
</xs:simpleType>

```

```

<xs:element name="IESTM33-22">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="Version" type="xs:string" fixed="1.1"/>
      <xs:element name="Header">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="Manufacturer" type="xs:string"
              minOccurs="0"/>
            <xs:element name="CatalogNumber" type="xs:string"
              minOccurs="0"/>
            <xs:element name="GTIN" type="xs:integer"
              minOccurs="0"/>
            <xs:element name="Description" type="xs:string"/>
            <xs:element name="Laboratory" type="xs:string"/>
            <xs:element name="ReportNumber" type="xs:string"/>
            <xs:element name="ReportDate" type="xs:date"/>
            <xs:element name="DocumentCreator" type="xs:string"
              minOccurs="0"/>
            <xs:element name="DocumentCreationDate" type="xs:date"
              minOccurs="0"/>
            <xs:element name="UniqueIdentifier" type="xs:string"
              minOccurs="0"/>
            <xs:element name="Comment" type="xs:string"
              minOccurs="0" maxOccurs="unbounded"/>
            <xs:element name="Reference" type="xs:string"
              minOccurs="0" maxOccurs="unbounded"/>
            <xs:element name="MoreInfoURI" type="xs:anyURI"
              minOccurs="0"/>
          </xs:sequence>
        </xs:complexType>
      </xs:element>
      <xs:element name="Luminaire" minOccurs="0">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="Dimensions">
              <xs:complexType>
                <xs:sequence>
                  <xs:element ref="Length"/>
                  <xs:element ref="Width"/>
                  <xs:element ref="Height"/>
                </xs:sequence>
              </xs:complexType>
            </xs:element>
            <xs:element name="Shape" minOccurs="0">
              <xs:simpleType>
                <xs:restriction base="xs:string">

```

```

        <xs:enumeration value="Align _X"/>
        <xs:enumeration value="Align _Y"/>
        <xs:enumeration value="Align _Z"/>
    </xs:restriction>
</xs:simpleType>
</xs:element>
    <xs:element name="NumEmitter" type="xs:int"/>
</xs:sequence>
</xs:complexType>
</xs:element>
<xs:element name="Equipment" minOccurs="0">
    <xs:complexType>
        <xs:sequence>
            <xs:element name="Gonioradiometer" minOccurs="0">
                <xs:complexType>
                    <xs:sequence>
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                                    <xs:enumeration value="CIE _B"/>
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                        <xs:element name="Rfh11" type="xs:int"
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        <xs:element ref="NumberVert"/>
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      <xs:element ref="NumberVert"/>
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            <xs:element name="Normalized"
              type="xs:boolean" minOccurs="0"/>
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              <xs:attribute ref="z"/>
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```

```

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                                    <xs:element ref="WidthOffset"/>
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        type="RegulatoryValue" minOccurs="0"/>
      <xs:element name="ColorTemperature"
        type="RegulatoryValue" minOccurs="0"/>
      <xs:element name="CIE _ CRI"
        type="RegulatoryValue" minOccurs="0"/>
      <xs:element name="IES _ TM30"
        type="RegulatoryValue" minOccurs="0"/>
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        type="RegulatoryValue" minOccurs="0"/>
      <xs:element name="SPRatio"
        type="RegulatoryValue" minOccurs="0"/>
      <xs:element name="LuminousIntensity"
        type="RegulatoryValue" minOccurs="0"/>
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        type="RegulatoryValue" minOccurs="0"/>
      <xs:element name="RadiantIntensity"
        type="RegulatoryValue" minOccurs="0"/>
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        type="RegulatoryValue" minOccurs="0"/>
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        type="RegulatoryValue" minOccurs="0"/>
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        type="RegulatoryValue" minOccurs="0"/>
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        type="RegulatoryValue" minOccurs="0"/>
      <xs:element name="AngularColor"
        type="RegulatoryValue" minOccurs="0"/>
      <xs:element name="Illuminance"
        type="RegulatoryValue" minOccurs="0"/>
      <xs:element name="Irradiance"
        type="RegulatoryValue" minOccurs="0"/>
      <xs:element name="PhotonFluxDensity"
        type="RegulatoryValue" minOccurs="0"/>
      <xs:element name="SpectralIrradiance"
        type="RegulatoryValue" minOccurs="0"/>
    </xs:sequence>
  </xs:complexType>
</xs:element>

```

```

        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
</xs:element>
<xs:element name="CustomData" minOccurs="0"
  maxOccurs="unbounded">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="Name" type="xs:string"/>
      <xs:element name="UniqueIdentifier" type="xs:string"/>
      <xs:any namespace="##any" processContents="skip"
        minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
  </xs:complexType>
</xs:element>
</xs:sequence>
</xs:complexType>
</xs:element>
</xs:schema>

```

4.8 JSON Document Format

JavaScript Object Notation (JSON) is a minimalist data-interchange format that is widely used for Internet client-server communications. Unlike the rich semantics of XML with its many data types, JSON allows only name-value pairs and ordered lists.

The advantage of the XML document format in representing luminaire optical data is that the documents can be validated using the associated schema (see **Section 4.7**); the advantage of the JSON document format is that its ordered lists can be used to compactly represent data arrays, such as luminous intensity and photon flux density. In particular, JSON documents tend to be an order of magnitude smaller than their equivalent XML documents.

XML- and JSON-formatted versions of this standard shall be interchangeable, such that one version can be converted into the other version and back again without any loss of information.

4.8.1 XML to JSON Mapping Mapping XML elements to JSON name-value pairs is straightforward. For example, the XML complex element:

```

<Equipment>
  <Gonioradiometer>
    <Type>CIE _ C</Type>
  </Gonioradiometer>
</Equipment>

```

... becomes this in JSON:

```

"Equipment": {
  "Gonioradiometer": {
    "Type": "CIE _ C"
  }
}

```

Where multiple XML elements are allowed, these can be represented in JSON as arrays. For example, this:

```
<Comment>This is a comment</Comment>
<Comment>This is another comment</Comment>
... becomes this:
"Comments": {
  "This is a comment",
  "This is another comment"
}
```

Elements without attributes are described in **Section 4.8.1.1**. Those with attributes are described in **Section 4.8.1.2**.

4.8.1.1 Multiple XML Elements Without Attributes The XML elements without attributes that can be represented as JSON arrays are shown in **Table 61**.

Table 61. Multiple XML Elements Without Attributes

Section	Element Name
4.2.11	Comment
4.2.12	Reference
4.4.1.2	MeasurementEquipment
4.4.2.1	MeasurementEquipment
4.4.3.1	MeasurementEquipment
4.5	Emitter
4.5.12.2.4	MeasUncertainty
4.5.17.1	EmitterSpectral
4.5.18.2	PlaneData
4.5.19.4	PlaneData
4.5.20.4	PlaneData
4.5.23.1.2	TopArea
4.5.23.2.2	BottomArea
4.5.23.3.2	C0Area
4.5.23.4.2	C90Area
4.5.23.5.2	C180Area
4.5.23.6.2	C270Area
4.5.26	Regulatory
4.6	CustomData

4.8.1.2 Multiple XML Elements With Attributes The XML elements with attributes that can be represented as JSON arrays are shown in **Table 62**.

Table 62. Multiple XML Elements With Attributes

Section	Element Names
4.5.8	<i>Tilt</i>
4.5.14.1	<i>Intensity Data</i>
4.5.15.3	<i>Intensity Data</i>
4.5.16.3	<i>Intensity Data</i>
4.5.17.2.8	<i>Intensity Data</i>
4.5.17.1	<i>Power Data</i>

(continued on next page)

Table 62. Multiple XML Elements With Attributes (continued)

Section	Element Names
4.5.17.2	<i>Angular Spectral Intensity</i>
4.5.18	<i>Angular Color</i>
4.5.19.4.1	<i>Plane Normal</i>
4.5.20.6.1	<i>Plane Normal</i>
4.5.21.6.1	<i>Plane Normal</i>
4.5.22.4.1	<i>Plane Normal</i>
4.5.19	<i>Illuminance Data</i>
4.5.20	<i>Irradiance Data</i>
4.5.21	<i>Photon Flux Density Data</i>
4.5.22	<i>Spectral Irradiance Data</i>
4.5.23	<i>Channels</i>
4.5.26	<i>Regulatory</i>
4.6	<i>CustomData</i>

4.8.1.2.1 Tilt Element A typical Tilt Angles XML element (see **Section 4.5.8**) is:

```
<TiltAngles>
  <NumberAngles>3</NumberAngles>
  <Tilt angle="0.0">1.00</Tilt>
  <Tilt angle="10.0">0.90</Tilt>
  <Tilt angle="20.0">0.80</Tilt>
</TiltAngles>
```

The equivalent JSON elements are:

```
"TiltAngles": {
  "NumberAngles": 3,
  "TiltArray": {
    "Angle": [0.0, 10.0, 20.0],
    "Mult": [1.00, 0.90, 0.80]
  }
}
```

4.8.1.2.2 Intensity Data Element A typical Luminous Intensity Data XML element (see **Section 4.5.14.1**) is:

```
<LuminousIntensity>
  <AbsolutePhotometry>true</AbsolutePhotometry>
  <Symm>Symm_Full</Symm>
  <NumberMeasured>95</NumberMeasured>
  <NumberHorz>5</NumberHorz>
  <NumberVert>19</NumberVert>
  ...
  <IntData h="0.0" v="60.0">53</IntData>
  <IntData h="0.0" v="65.0">44</IntData>
  ...
</LuminousIntensity>
```

If the `Symm` type is `Symm_Arbitrary` and `NumberHorz` is zero and `NumberVert` is zero (that is, the measurement coordinate system does not consist of a set of planes), the equivalent JSON elements are:

```
"LuminousIntensity": {
  "AbsolutePhotometry": true,
  "Symm": "Symm_Arbitrary",
  "NumberMeasured": 95,
  "NumberHorz": 0,
  "NumberVert": 0,
  "IntDataNoSymm": [
    ...
    [0.0, 60.0, 53],
    [5.0, 65.0, 44],
    ...
  ]
}
```

Otherwise, horizontal symmetry allows the intensity data to be expressed more compactly as:

```
"LuminousIntensity": {
  "AbsolutePhotometry": true,
  "Symm": "Symm_Full",
  "NumberMeasured": 95,
  "NumberHorz": 5,
  "NumberVert": 19,
  "IntDataSymm": {
    "h": [0.0],
    "v": [ ..., 60.0, 65.0, ... ],
    "IntData": [
      [ ..., 53, 44, ... ],
      ...
      [ ..., 50, 43, ... ]
    ]
  }
}
```

... where “IntData” is a two-dimensional array with “NumberHorz” rows and “NumberVert” columns. That is, it is ordered by “NumberHorz” and “NumberVert”.

Radiant intensity and photon intensity data elements (see **Sections 4.5.15.3** and **4.5.16.3**) can be similarly expressed.

4.8.1.2.3 Power Data Element A typical Emitter Spectral XML element (see **Section 4.5.17.1**) is:

```
<EmitterSpectral>
  <NumberWavelength>61</NumberWavelength>
  ...
  <PwrData w="555.0">0.053</PwrData>
  ...
</EmitterSpectral>
```

The equivalent JSON elements are:

```
"EmitterSpectral": {
  "NumberWavelength": 61,
  "PwrDataArray": {
    "w": [..., 555.0, ...],
    "PwrData": [..., 0.053, ...]
  }
}
```

4.8.1.2.4 Angular Spectral Intensity Data Element A typical Angular Spectral Intensity Data XML element (see **Section 4.5.17.2**) is:

```
<AngularSpectral>
  <Symm>Symm _ Full</Symm>
  <NumberMeasured>19</NumberMeasured>
  <NumberHorz>1</NumberHorz>
  <NumberVert>19</NumberVert>
  <NumberWavelength>81</NumberWavelength>
  ...
  <IntData h="0.0" v="60.0" w="555.0">0.023</IntData>
  <IntData h="0.0" v="60.0" w="560.0">0.031</IntData>
  ...
</AngularSpectral>
```

If the Symm type is Symm_Arbitrary and NumberHorz is zero and NumberVert is zero (that is, the measurement coordinate system does not consist of a set of planes), the equivalent JSON elements are:

```
"AngularSpectral": {
  "Symm": "Symm _ Arbitrary",
  "NumberMeasured": 19,
  "NumberHorz": 0,
  "NumberVert": 0,
  "NumberWavelength": 81,
  "IntDataNoSymm": [
    ...
    [0.0, 60.0, 555.0, 0.023],
    [0.0, 60.0, 555.0, 0.031],
    ...
  ]
}
```

Otherwise, horizontal symmetry allows the intensity data to be expressed more compactly as:

```
"AngularSpectral": {
  "Symm": "Symm _ Full",
  "NumberMeasured": 19,
  "NumberHorz": 1,
  "NumberVert": 19,
  "NumberWavelength": 81,
  "IntDataSymm": {
```

```

    "h": [0.0],
    "v": [..., 60.0, ...],
    "w": [..., 555.0, 560.0, ...],
    "IntData": [
        [
            ...
            [ ..., 0.023, 0.031, ... ],
            ...
        ]
    ]
}

```

... where “IntData” is a three-dimensional array ordered by “NumberHorz”, “NumberVert”, and “NumberWavelength”.

4.8.1.2.5 Angular Color Data Element A typical Angular Color Data XML element (see Section 4.5.18) is:

```

<AngularColor>
  <Symm>Symm_Full</Symm>
  <NumberMeasured>19</NumberMeasured>
  <NumberHorz>1</NumberHorz>
  <NumberVert>19</NumberVert>
  ...
  <ColorData h="0.0" v="60.0" x="0.3135" y="0.3293">55</ColorData>
  <ColorData h="0.0" v="65.0" y="0.3129" y="0.3290">52</ColorData>
  ...
</AngularColor>

```

If the Symm type is Symm_Arbitrary and NumberHorz is zero and NumberVert is zero (that is, the measurement coordinate system does not consist of a set of planes), the equivalent JSON elements are:

```

"AngularColor": {
  "Symm": "Symm_Arbitrary",
  "NumberMeasured": 19,
  "NumberHorz": 0,
  "NumberVert": 0,
  "NumberWavelength": 81,
  "ColorDataNoSymm": [
    ...
    [0.0, 60.0, 0.3135, 0.3293, 55],
    [0.0, 65.0, 0.3129, 0.3290, 52],
    ...
  ]
}

```

Otherwise, horizontal symmetry allows the color data to be expressed more compactly as:

```

"AngularColor": {
  "Symm": "Symm_Full",
  "NumberMeasured": 19,

```

```

    "NumberHorz": 1,
    "NumberVert": 19,
    "ColorDataSymm": {
        "h": [0.0],
        "v": [..., 60.0, 65.0, ...],
        "ColorData": [
            [
                ...
                [0.3135, 0.3293, 55],
                [0.3129, 0.3290, 52],
                ...
            ]
        ]
    }
}

```

... where "IntData" is a two-dimensional array ordered by "NumberHorz" and "NumberVert".

4.8.1.2.6 Plane Normal Element A typical Plane Normal XML element (see **Sections 4.5.19.4.1, 4.5.20.6.1, 4.5.21.6.1, and 4.5.22.4.1**) is:

```
<PlaneNormal x="1.0" y="0.0" z="0.0"/>
```

The equivalent JSON element is:

```
"PlaneNormal": [1.0,0.0,0.0]
```

4.8.1.2.7 Illuminance Data Element A typical Illuminance Data element (see **Section 4.5.19**) is:

```

<IllumData>
  <NumberPlanes>1</NumberPlanes>
  <PlaneData>
    <NumberMeasured>128</NumberMeasured>
    ...
    <Illum x="0.0" y="1.5" z="1.2">55</Illum>
    ...
  </PlaneData>
</IllumData>

```

If the illuminance measurements are not on a rectangular grid, the equivalent JSON elements are:

```

"IllumData": {
  "NumberPlanes": 1,
  "PlaneData": [
    "NumberMeasured": 128,
    "IllumNoSymm": [
      ...
      [0.0,1.5,1.2,55],
      ...
    ]
  ]
}

```

```
}
```

Otherwise, symmetry allows the illuminance data to be expressed more compactly as:

```
"IllumData": {
  "NumberPlanes": 1,
  "PlaneData": [
    "IllumSymm": {
      "x": [..., 0.0, ...],
      "y": [..., 1.5, ...],
      "z": [..., 1.2, ...],
      "Illum": [
        [
          ...
          [..., 55, ...]
          ...
        ]
      ]
    }
  ]
}
```

... where "Illum" is a three-dimensional array ordered by "x", "y", and "z".

4.8.1.2.7 Illuminance Data Element A typical Illuminance Data element (see **Section 4.5.19**) is:

```
<IllumData>
  <NumberPlanes>1</NumberPlanes>
  <PlaneData>
    <NumberMeasured>128</NumberMeasured>
    ...
    <Illum x="0.0" y="1.5" z="1.2">55</Illum>
    ...
  </PlaneData>
</IllumData>
```

If the illuminance measurements are not on a rectangular grid, the equivalent JSON elements are:

```
"IllumData": {
  "NumberPlanes": 1,
  "PlaneData": [
    "NumberMeasured": 128,
    "IllumNoSymm": [
      ...
      [0.0,1.5,1.2,55],
      ...
    ]
  ]
}
```

Otherwise, symmetry allows the illuminance data to be expressed more compactly as:

```
"IllumData": {
  "NumberPlanes": 1,
  "PlaneData": [
    "IllumSymm": {
      "x": [..., 0.0, ...],
      "y": [..., 1.5, ...],
      "z": [..., 1.2, ...],
      "Illum": [
        [
          ...
          [..., 55, ...]
          ...
        ]
      ]
    }
  ]
}
```

... where "Illum" is a three-dimensional array ordered by "x", "y", and "z".

4.8.1.2.8 Irradiance Data Element A typical Irradiance Data element (see Section 4.5.20) is:

```
<IrradData>
  <NumberPlanes>1</NumberPlanes>
  <PlaneData>
    <NumberMeasured>128</NumberMeasured>
    ...
    <Irrad x="0.0" y="1.5" z="1.2">55</Irrad>
    ...
  </PlaneData>
</IrradData>
```

If the irradiance measurements are not on a rectangular grid, the equivalent JSON elements are:

```
"IrradData": {
  "NumberPlanes": 1,
  "PlaneData": [
    "NumberMeasured": 128,
    "IrradNoSymm": [
      ...
      [0.0,1.5,1.2,55],
      ...
    ]
  ]
}
```

Otherwise, symmetry allows the irradiance data to be expressed more compactly as:

```
"IrradData": {
  "NumberPlanes": 1,
  "PlaneData": [
    "IrradSymm": {
      "x": [..., 0.0, ...],
      "y": [..., 1.5, ...],
      "z": [..., 1.2, ...],
      "Irrad": [
        [
          ...
          [..., 55, ...]
          ...
        ]
      ]
    }
  ]
}
```

... where "Irrad" is a three-dimensional array ordered by "x", "y", and "z".

4.8.1.2.9 Photon Flux Density Data Element A typical Photon Flux Density Data element (see **Section 4.5.21**) is:

```
<PFDData>
  <NumberPlanes>1</NumberPlanes>
  <PlaneData>
    <NumberMeasured>128</NumberMeasured>
    ...
    <PFD x="0.0" y="1.5" z="1.2">55</PFD>
    ...
  </PlaneData>
</PFDData>
```

If the photon flux density measurements are not on a rectangular grid, the equivalent JSON elements are:

```
"PFDData": {
  "NumberPlanes": 1,
  "PlaneData": [
    "NumberMeasured": 128,
    "PFDDNoSymm": [
      ...
      [0.0,1.5,1.2,55],
      ...
    ]
  ]
}
```

Otherwise, symmetry allows the photon flux density data to be expressed more compactly as:

```
"PFDData": {
  "NumberPlanes": 1,
```

```

"PlaneData": [
  "PFDSymm": {
    "x": [..., 0.0, ...],
    "y": [..., 1.5, ...],
    "z": [..., 1.2, ...],
    "PFD": [
      [
        ...
        [..., 55, ...]
        ...
      ]
    ]
  }
]
}

```

... where "PFD" is a three-dimensional array ordered by "x", "y", and "z".

4.8.1.2.10 Spectral Irradiance Data Element A typical Spectral Irradiance Data element (see **Section 4.5.22**) is:

```

<SpecIrradData>
  <NumberPlanes>1</NumberPlanes>
  <PlaneData>
    <NumberMeasured>128</NumberMeasured>
    ...
    <SIrrad x="0.0" y="1.5" z="1.2" w="555.0">55</SIrrad>
    ...
  </PlaneData>
</SpecIrradData>

```

If the irradiance measurements are not on a rectangular grid, the equivalent JSON elements are:

```

"SpecIrradData": {
  "NumberPlanes": 1,
  "PlaneData": [
    "NumberMeasured": 128,
    "SpecIrradNoSymm": [
      ...
      [0.0, 1.5, 1.2, 555.0, 55],
      ...
    ]
  ]
}

```

Otherwise, symmetry allows the irradiance data to be expressed more compactly as:

```

"SpecIrradData": {
  "NumberPlanes": 1,
  "PlaneData": [
    "SpecIrradSymm": {

```

```

"x": [..., 0.0, ...],
"y": [..., 1.5, ...],
"z": [..., 1.2, ...],
"w": [..., 555.0, ...],
"SIrrad": [
  [
    [
      ...
      [..., 55, ...]
      ...
    ]
  ]
]
}
]
}

```

... where "SIrrad" is a four-dimensional array ordered by "x", "y", "z", and "w".

4.8.1.11 Channel Element A typical Channel element (see **Section 4.5.23**) is:

```

<Channels>
  <NumChannels>3</NumChannels>
  <ChannelMult name="red">0.21</ChannelMult>
  <ChannelMult name="green">0.72</ChannelMult>
  <ChannelMult name="blue">0.08</ChannelMult>
</Channels>

```

The equivalent JSON elements are:

```

"Channels": {
  "NumChannels": 3,
  "ChannelMult": [
    { "Name"="red", "ChannelMult"=0.21 },
    { "Name"="green", "ChannelMult"=0.72 },
    { "Name"="blue", "ChannelMult"=0.08 }
  ]
}

```

4.8.2 JSON Schema The JSON schema follows:

```

{
  "$schema": "http://json-schema.org/draft/2019-09/schema#",
  "type": "object",
  "properties": {
    "FileType": {
      "type": "string"
    }
  }
  "Version": {
    "type": "number"
  }
}

```

```

},
"Header": {
  "type": "object",
  "properties": {
    "Manufacturer": {
      "type": "string"
    },
    "CatalogNumber": {
      "type": "string"
    },
    "GTIN": {
      "type": "number"
    },
    "Description": {
      "type": "string"
    },
    "Laboratory": {
      "type": "string"
    },
    "ReportNumber": {
      "type": "string"
    },
    "ReportDate": {
      "type": "string",
      "pattern": "^[0-9]{4}-[0-9]{2}-[0-9]{2})$|UNKNOWN)"
    },
    "DocumentCreator": {
      "type": "string"
    },
    "DocumentCreationDate": {
      "type": "string",
      "pattern": "^[0-9]{4}-[0-9]{2}-[0-9]{2})$"
    },
    "UniqueIdentifier": {
      "type": "string"
    },
    "Comments": {
      "type": "array",
      "items": {
        "type": "string"
      }
    },
    "Reference": {
      "type": "array",
      "items": {
        "type": "string"
      }
    }
  }
}

```

```

    },
    "MoreInfoURI": {
        "type": "string"
    }
},
"required": [
    "Description", "Laboratory", "ReportNumber", "ReportDate"
]
},
"Luminaire": {
    "type": "object",
    "properties": {
        "Dimensions": {
            "type": "object",
            "properties": {
                "Length": {
                    "type": "number"
                },
                "Width": {
                    "type": "number"
                },
                "Height": {
                    "type": "number"
                }
            }
        },
        "required": ["Length", "Width", "Height"]
    },
    "Shape": {
        "type": "string",
        "enum": ["Align_X", "Align_Y", "Align_Z"]
    },
    "NumEmitter": {
        "type": "integer"
    }
},
"required": ["Dimensions", "NumEmitter"]
},
"Equipment": {
    "type": "object",
    "properties": {
        "Gonioradiometer": {
            "type": "array",
            "items": {
                "type": "object",
                "properties": {
                    "Type": {
                        "type": "string",

```

```

        "enum": [
            "CIE _ A",
            "CIE _ B",
            "CIE _ C",
            "IES _ A",
            "IES _ B",
            "IES _ C",
            "CUSTOM"
        ]
    },
    "MeasurementEquipment": {
        "type": "array",
        "items": {
            "type": "string"
        }
    }
},
"required": ["Type"]
},
"IntegratingSphere": {
    "type": "array",
    "items": {
        "MeasurementEquipment": {
            "type": "string"
        }
    }
},
"required": ["MeasurementEquipment"]
},
"Spectroradiometer": {
    "type": "array",
    "items": {
        "type": "object",
        "properties": {
            "MeasurementEquipment": {
                "type": "string"
            },
            "BandwidthFWHM": {
                "type": "number"
            },
            "BandwidthCorrected": {
                "type": "boolean"
            },
            "BandwidthMethod": {
                "type": "string"
            }
        },
        "required": ["MeasurementEquipment"]
    }
}

```

```

    },
    "required": ["MeasurementEquipment"]
  }
}
},
"Emitter": {
  "type": "array",
  "items": {
    "type": "object",
    "properties": {
      "Quantity": {
        "type": "integer"
      },
      "Description": {
        "type": "string"
      },
      "CatalogNumber": {
        "type": "string"
      },
      "RatedLumens": {
        "type": "number"
      },
      "InputWattage": {
        "type": "number"
      },
      "PowerFactor": {
        "type": "number"
      },
      "BallastFactor": {
        "type": "number"
      },
      "TiltAngles": {
        "type": "object",
        "properties": {
          "NumberAngles": {
            "type": "integer"
          },
          "TiltArray": {
            "type": "object",
            "properties": {
              "Angle": {
                "type": "array",
                "items": {
                  "type": "number"
                }
              }
            }
          }
        }
      },
    },
  },
}

```

```

        "Mult": {
            "type": "array",
            "items": {
                "type": "number"
            }
        },
        "required": ["Angle", "Mult"]
    },
    "required": ["NumberAngles", "TiltArray"]
},
"ColorTemperature": {
    "type": "object",
    "properties": {
        "FixedCCT": {
            "type": "integer"
        },
        "MinCCT": {
            "type": "integer"
        },
        "MaxCCT": {
            "type": "integer"
        }
    }
},
"ColorRendering": {
    "type": "object",
    "properties": {
        "CIE_CRI": {
            "type": "object",
            "properties": {
                "Ra": {
                    "type": "integer"
                },
                "R9": {
                    "type": "integer"
                }
            }
        },
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    },
    "IES_TM30": {
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        "properties": {
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        },

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},
"Rfh02": {
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"Rfh03": {
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},
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  "type": "integer"
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"Rfh15": {
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}
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"Rcsh02": {
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"Rcsh03": {
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"Rcsh07": {
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"Rcsh14": {
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"Rcsh15": {

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        "Simulation": {
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            "properties": {
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                    "enum": [
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                "ApprovalScope": {
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                    "items": {
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                            "Uncertainty": {
                                "type": "number"
                            }
                        }
                    }
                }
            }
        }
    }
}

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        }
    }
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}
},
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        "ApprovalBody",
        "ApprovalScope",
        "MeasUncertainty"
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    "AngleInterpolation": {
        "type": "boolean"
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            "properties": {
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                        "Symm _ Bi _ 0",
                        "Symm _ Bi _ 90",
                        "Symm _ Quad",
                        "Symm _ Full",
                        "Symm _ Arbitrary"
                    ]
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                "NumberMeasured": {
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            }
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    }
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    "NumberVert": {
      "type": "integer"
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    "IntDataSymm": {
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      "properties": {
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          "items": {
            "type": "number"
          }
        },
        "v": {
          "type": "array",
          "items": {
            "type": "number"
          }
        },
        "IntData": {
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          "items": {
            "type": "array",
            "items": {
              "type": "number"
            }
          }
        }
      }
    },
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  "IntDataNoSymm": {
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        "type": "number",
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        "maxItems": 3
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  }
},
"oneOf": [
  {

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            "NumberHorz",
            "NumberVert",
            "IntDataSymm"
        ]
    },
    {
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            "NumberMeasured",
            "NumberHorz",
            "NumberVert",
            "IntDataNoSymm"
        ]
    }
],
    },
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    "required": ["LuminousIntensity"]
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        "properties": {
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            "MaxWavelength": {
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            "RadiantIntensity": {
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                "properties": {
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                        "type": "boolean"
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                    "Symm": {
                        "type": "string",
                        "enum": [
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                            "Symm _ Bi _ 0",
                            "Symm _ Bi _ 90",
                            "Symm _ Quad",
                            "Symm _ Full",
                            "Symm _ Arbitrary"
                        ]
                    }
                }
            }
        }
    }
}

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```

    ]
  },
  "Multiplier": {
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  "NumberMeasured": {
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  },
  "NumberHorz": {
    "type": "integer"
  },
  "NumberVert": {
    "type": "integer"
  },
  "IntDataSymm": {
    "type": "object",
    "properties": {
      "h": {
        "type": "array",
        "items": {
          "type": "number"
        }
      },
      "v": {
        "type": "array",
        "items": {
          "type": "number"
        }
      },
      "IntData": {
        "type": "array",
        "items": {
          "type": "array",
          "items": {
            "type": "number"
          }
        }
      }
    }
  },
  "required": ["h", "v", "IntData"]
},
"IntDataNoSymm": {
  "type": "array",
  "items": {
    "type": "array",
    "items": {
      "type": "number",

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        "minItems": 3,
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    }
  },
  "oneOf": [
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        "NumberHorz",
        "NumberVert",
        "IntDataSymm"
      ]
    },
    {
      "required": [
        "NumberMeasured",
        "NumberHorz",
        "NumberVert",
        "IntDataNoSymm"
      ]
    }
  ],
  "RadiantFlux": {
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  ],
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      "MaxWavelength": {
        "type": "number"
      },
      "PhotonIntensity": {
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        "properties": {
          "Absolute": {
            "type": "boolean"
          }
        }
      }
    }
  }
}

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},
"Symm": {
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    "Symm _ Bi _ 90",
    "Symm _ Quad",
    "Symm _ Full",
    "Symm _ Arbitrary"
  ]
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},
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  "type": "integer"
},
"NumberHorz": {
  "type": "integer"
},
"NumberVert": {
  "type": "integer"
},
"IntDataSymm": {
  "type": "object",
  "properties": {
    "h": {
      "type": "array",
      "items": {
        "type": "number"
      }
    },
    "v": {
      "type": "array",
      "items": {
        "type": "number"
      }
    }
  },
  "IntData": {
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    "items": {
      "type": "array",
      "items": {
        "type": "number"
      }
    }
  }
}

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        }
    },
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},
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        "items": {
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            "maxItems": 3
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}
},
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        "required": [
            "NumberMeasured",
            "NumberHorz",
            "NumberVert",
            "IntDataSymm"
        ]
    },
    {
        "required": [
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            "NumberHorz",
            "NumberVert",
            "IntDataNoSymm"
        ]
    }
]
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"PhotonFlux": {
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},
"required": [
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]
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"SpectralData": {
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    "properties": {
        "EmitterSpectral": {

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    "properties": {
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        "type": "string"
      },
      "NumberWavelength": {
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      "Multiplier": {
        "type": "number"
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      "Normalized": {
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      },
      "TimeStamp": {
        "type": "integer"
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      "PwrDataArray": {
        "type": "object",
        "properties": {
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            "type": "array",
            "items": {
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            }
          },
          "PwrData": {
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            "items": {
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            }
          }
        },
        "required": ["w", "PwrData"]
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      "Quantum": {
        "type": "boolean"
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    "required": ["NumberWavelength", "PwrDataArray"]
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  "AngularSpectral": {
    "type": "object",
    "properties": {
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        "type": "boolean"
      }
    },

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"Symm": {
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    "Symm _ Bi _ 90",
    "Symm _ Quad",
    "Symm _ Full",
    "Symm _ Arbitrary"
  ]
},
"Multiplier": {
  "type": "number"
},
"NumberMeasured": {
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},
"NumberHorz": {
  "type": "integer"
},
"NumberVert": {
  "type": "integer"
},
"NumberWavelength": {
  "type": "integer"
},
"IntDataSymm": {
  "type": "object",
  "properties": {
    "h": {
      "type": "array",
      "items": {
        "type": "number"
      }
    },
    "v": {
      "type": "array",
      "items": {
        "type": "number"
      }
    },
    "w": {
      "type": "array",
      "items": {
        "type": "number"
      }
    }
  }
},

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                    "items": {
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                    }
                }
            }
        },
        "required": ["h", "v", "w", "IntData"]
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    "IntDataNoSymm": {
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        "items": {
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            "items": {
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                "minItems": 4,
                "maxItems": 4
            }
        }
    },
    "oneOf": [
        {
            "required": [
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                "NumberHorz",
                "NumberVert",
                "NumberWavelength",
                "IntDataSymm"
            ]
        },
        {
            "required": [
                "NumberMeasured",
                "NumberHorz",
                "NumberVert",
                "NumberWavelength",
                "IntDataNoSymm"
            ]
        }
    ]
}

```

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    }
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        "enum": [
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          "Symm _ Bi _ 0",
          "Symm _ Bi _ 90",
          "Symm _ Quad",
          "Symm _ Full",
          "Symm _ Arbitrary"
        ]
      },
    },
    "Multiplier": {
      "type": "number"
    },
    "NumberMeasured": {
      "type": "integer"
    },
    "NumberHorz": {
      "type": "integer"
    },
    "NumberVert": {
      "type": "integer"
    },
    "ColorDataSymm": {
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      "properties": {
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          "type": "array",
          "items": {
            "type": "number"
          }
        },
        "v": {
          "type": "array",
          "items": {
            "type": "number"
          }
        },
      },
      "ColorData": {

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        "type": "array",
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            "items": {
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            "minItems": 3,
            "maxItems": 3
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    "items": {
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        "items": {
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            "maxItems": 5
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    }
}
},
"oneOf": [
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        "required": [
            "NumberMeasured",
            "NumberHorz",
            "NumberVert",
            "ColorDataSymm"
        ]
    },
    {
        "required": [
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            "NumberHorz",
            "NumberVert",
            "ColorDataNoSymm"
        ]
    }
]
},
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    "properties": {

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"Multiplier": {
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},
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      "items": {
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      "minItems": 3,
      "maxItems": 3
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    "IllumSymm": {
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      "properties": {
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          "items": {
            "type": "number"
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          "items": {
            "type": "number"
          }
        },
        "z": {
          "type": "array",
          "items": {
            "type": "number"
          }
        },
        "Illum": {
          "type": "array",
          "items": {

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            "items": {
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            }
        }
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    },
    {
        "required": ["NumberMeasured", "IllumNoSymm"]
    }
]
},
"required": ["NumberPlanes", "PlaneData"]
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        },
        "Multiplier": {
            "type": "number"
        },
        "MinWavelength": {
            "type": "number"
        }
    }
}

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},
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},
"PlaneData": {
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  "properties": {
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      "items": {
        "type": "number"
      },
      "minItems": 3,
      "maxItems": 3
    },
    "NumberMeasured": {
      "type": "integer"
    },
    "IrradSymm": {
      "type": "object",
      "properties": {
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          "type": "array",
          "items": {
            "type": "number"
          }
        },
        "y": {
          "type": "array",
          "items": {
            "type": "number"
          }
        },
        "z": {
          "type": "array",
          "items": {
            "type": "number"
          }
        },
        "Irrad": {
          "type": "array",
          "items": {
            "type": "array",
            "items": {

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```

        "type": "array",
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        }
    }
}
},
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},
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    "items": {
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            "maxItems": 4
        }
    }
}
},
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    },
    {
        "required": ["NumberMeasured", "IrradNoSymm"]
    }
]
}
},
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    "MaxWavelength",
    "NumberPlanes",
    "PlaneData"
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},
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    "properties": {
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            "type": "boolean"
        },
        "Multiplier": {
            "type": "number"
        }
    }
}

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```

    },
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      "type": "number"
    },
    "MaxWavelength": {
      "type": "number"
    },
    "NumberPlanes": {
      "type": "integer"
    },
    "PlaneData": {
      "type": "object",
      "properties": {
        "PlaneNormal": {
          "type": "array",
          "items": {
            "type": "number"
          },
          "minItems": 3,
          "maxItems": 3
        },
        "NumberMeasured": {
          "type": "integer"
        },
        "PFDSymm": {
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          "properties": {
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              "type": "array",
              "items": {
                "type": "number"
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                "type": "number"
              }
            },
            "z": {
              "type": "array",
              "items": {
                "type": "number"
              }
            },
            "PFD": {
              "type": "array",

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```

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            "items": {
                "type": "array",
                "items": {
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                }
            }
        }
    },
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    "items": {
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        "items": {
            "type": "number",
            "minItems": 4,
            "maxItems": 4
        }
    }
}
},
"oneOf": [
    {
        "required": ["NumberMeasured", "PFDSymm"]
    },
    {
        "required": ["NumberMeasured", "PFDNoSymm"]
    }
]
}
},
"required": [
    "MinWavelength",
    "MaxWavelength",
    "NumberPlanes",
    "PlaneData"
]
},
"SpecIrradData": {
    "type": "object",
    "properties": {
        "Absolute": {
            "type": "boolean"
        }
    }
}

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```

},
"Multiplier": {
  "type": "number"
},
"NumberPlanes": {
  "type": "integer"
},
"PlaneData": {
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  "properties": {
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      "items": {
        "type": "number"
      },
      "minItems": 3,
      "maxItems": 3
    },
    "NumberMeasured": {
      "type": "integer"
    },
    "SpecIrradSymm": {
      "type": "object",
      "properties": {
        "x": {
          "type": "array",
          "items": {
            "type": "number"
          }
        },
        "y": {
          "type": "array",
          "items": {
            "type": "number"
          }
        },
        "z": {
          "type": "array",
          "items": {
            "type": "number"
          }
        },
        "w": {
          "type": "array",
          "items": {
            "type": "number"
          }
        }
      }
    }
  }
}

```

```

    },
    "SIrrad": {
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        "type": "array",
        "items": {
          "type": "array",
          "items": {
            "type": "array",
            "items": {
              "type": "number"
            }
          }
        }
      }
    }
  },
  "required": ["x", "y", "z", "SIrrad"]
},
"SpecIrradNoSymm": {
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  "items": {
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    "items": {
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      "minItems": 4,
      "maxItems": 4
    }
  }
}
}
},
"oneOf": [
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  },
  {
    "required": ["NumberMeasured", "SpecIrradNoSymm"]
  }
]
},
"required": [
  "MinWavelength",
  "MaxWavelength",
  "NumberPlanes",
  "PlaneData"
]

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```

    ]
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      "ChannelMult": {
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        "items": {
          "type": "object",
          "properties": {
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            },
            "ChannelMult": {
              "type": "number"
            }
          }
        },
        "required": ["name", "ChannelMult"]
      }
    }
  },
  "required": ["NumChannels", "ChannelMult"]
},
"EmissionAreas": {
  "type": "object",
  "properties": {
    "TopFace": {
      "type": "object",
      "properties": {
        "NumberTop": {
          "type": "integer"
        }
      },
      "TopArea": {
        "type": "object",
        "properties": {
          "Length": {
            "type": "number"
          },
          "Width": {
            "type": "number"
          },
          "LengthOffset": {
            "type": "number"
          }
        }
      }
    }
  }
}

```

```

        "WidthOffset": {
            "type": "number"
        },
        "Circular": {
            "type": "boolean"
        }
    },
    "required": [
        "Length",
        "Width",
        "LengthOffset",
        "WidthOffset"
    ]
}
},
"required": ["NumberTop", "TopArea"]
},
"BottomFace": {
    "type": "object",
    "properties": {
        "NumberBottom": {
            "type": "integer"
        },
        "BottomArea": {
            "type": "array",
            "items": {
                "type": "object",
                "properties": {
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                    },
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                        "type": "number"
                    },
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                    },
                    "Circular": {
                        "type": "boolean"
                    }
                }
            },
            "required": [
                "Length",
                "Width",

```

```

        "LengthOffset",
        "WidthOffset"
    ]
    }
}
},
"required": ["NumberBottom", "BottomArea"]
},
"C0Face": {
    "type": "object",
    "properties": {
        "NumberC0": {
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            "properties": {
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                },
                "Height": {
                    "type": "number"
                },
                "LengthOffset": {
                    "type": "number"
                },
                "HeightOffset": {
                    "type": "number"
                },
                "Circular": {
                    "type": "boolean"
                }
            }
        },
        "required": [
            "Length",
            "Height",
            "LengthOffset",
            "HeightOffset"
        ]
    }
},
"required": ["NumberC0", "C0Area"]
},
"C90Face": {
    "type": "object",
    "properties": {
        "NumberC90": {

```

```

        "type": "integer"
    },
    "C90Area": {
        "type": "object",
        "properties": {
            "Width": {
                "type": "number"
            },
            "Height": {
                "type": "number"
            },
            "WidthOffset": {
                "type": "number"
            },
            "HeightOffset": {
                "type": "number"
            },
            "Circular": {
                "type": "boolean"
            }
        }
    },
    "required": [
        "Width",
        "Height",
        "WidthOffset",
        "HeightOffset"
    ]
}

},
"required": ["NumberC90", "C90Area"]
},
"C180Face": {
    "type": "object",
    "properties": {
        "NumberC180": {
            "type": "integer"
        },
        "C180Area": {
            "type": "object",
            "properties": {
                "Length": {
                    "type": "number"
                },
                "Height": {
                    "type": "number"
                },
                "LengthOffset": {

```

```

        "type": "number"
    },
    "HeightOffset": {
        "type": "number"
    },
    "Circular": {
        "type": "boolean"
    }
},
"required": [
    "Length",
    "Height",
    "LengthOffset",
    "HeightOffset"
]
}
},
"required": ["NumberC180", "C180Area"]
},
"C270Face": {
    "type": "object",
    "properties": {
        "NumberC270": {
            "type": "integer"
        },
        "C270Area": {
            "type": "object",
            "properties": {
                "Width": {
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                },
                "Height": {
                    "type": "number"
                },
                "WidthOffset": {
                    "type": "number"
                },
                "HeightOffset": {
                    "type": "number"
                },
                "Circular": {
                    "type": "boolean"
                }
            }
        },
        "required": [
            "Width",
            "Height",

```

```

        "WidthOffset",
        "HeightOffset"
    ]
    }
    },
    "required": ["NumberC270", "C270Area"]
}
},
"EmitterCenter": {
    "type": "object",
    "properties": {
        "LengthOffset": {
            "type": "number"
        },
        "WidthOffset": {
            "type": "number"
        },
        "HeightOffset": {
            "type": "number"
        }
    },
    "required": [
        "LengthOffset", "WidthOffset", "HeightOffset"
    ]
},
"Regulatory": {
    "type": "object",
    "properties": {
        "InputWattage": {
            "type": "string",
            "enum": ["Measured", "Nominal", "Rated"]
        },
        "PowerFactor": {
            "type": "string",
            "enum": ["Measured", "Nominal", "Rated"]
        },
        "BallastFactor": {
            "type": "string",
            "enum": ["Measured", "Nominal", "Rated"]
        },
        "ColorTemperature": {
            "type": "string",
            "enum": ["Measured", "Nominal", "Rated"]
        },
        "CIE_CRI": {
            "type": "string",

```

```

        "enum": ["Measured", "Nominal", "Rated"]
    },
    "IES_TM30": {
        "type": "string",
        "enum": ["Measured", "Nominal", "Rated"]
    },
    "Duv": {
        "type": "string",
        "enum": ["Measured", "Nominal", "Rated"]
    },
    "SPRatio": {
        "type": "string",
        "enum": ["Measured", "Nominal", "Rated"]
    },
    "LuminousIntensity": {
        "type": "string",
        "enum": ["Measured", "Nominal", "Rated"]
    },
    "LuminousFlux": {
        "type": "string",
        "enum": ["Measured", "Nominal", "Rated"]
    },
    "RadiantIntensity": {
        "type": "string",
        "enum": ["Measured", "Nominal", "Rated"]
    },
    "RadiantFlux": {
        "type": "string",
        "enum": ["Measured", "Nominal", "Rated"]
    },
    "PhotonIntensity": {
        "type": "string",
        "enum": ["Measured", "Nominal", "Rated"]
    },
    "PhotonFlux": {
        "type": "string",
        "enum": ["Measured", "Nominal", "Rated"]
    },
    "SpectralPower": {
        "type": "string",
        "enum": ["Measured", "Nominal", "Rated"]
    },
    "SpectralIntensity": {
        "type": "string",
        "enum": ["Measured", "Nominal", "Rated"]
    },
    "AngularColor": {

```

```

        "type": "string",
        "enum": ["Measured", "Nominal", "Rated"]
    },
    "Illuminance": {
        "type": "string",
        "enum": ["Measured", "Nominal", "Rated"]
    },
    "Irradiance": {
        "type": "string",
        "enum": ["Measured", "Nominal", "Rated"]
    },
    "PhotonFluxDensity": {
        "type": "string",
        "enum": ["Measured", "Nominal", "Rated"]
    },
    "SpectralIrradiance": {
        "type": "string",
        "enum": ["Measured", "Nominal", "Rated"]
    }
}

},
"required": ["Quantity", "Description", "InputWattage"]
}

},
"CustomData": {
    "type": "array",
    "items": {
        "type": "object",
        "properties": {
            "Name": {
                "type": "string"
            },
            "UniqueIdentifier": {
                "type": "string"
            }
        },
        "required": ["Name", "UniqueIdentifier"],
        "additionalProperties": true
    }
}

},
"required": ["Header", "Emitter"]
}

```

Annex A – Sample Luminous (Photometric) Documents

A.1 Sample Luminous (Photometric) XML Document

```

<?xml version="1.1"?>
<IESTM33-22>
  <Version>1.0</Version>
  <Header>
    <Manufacturer>Academy Lighting</Manufacturer>
    <CatalogNumber>XET 55529</CatalogNumber>
    <Description>LED 2' x 4' Troffer</Description>
    <Laboratory>Apex Analytics</Laboratory>
    <ReportNumber>APEX-091101-004</ReportNumber>
    <ReportDate>2009-11-01</ReportDate>
    <DocumentCreator>Apex Analytics</DocumentCreator>
    <UniqueIdentifier>21EC2020-3AEA-4069-A2DD-08002B30309D
      </UniqueIdentifier>
    <Comment>Ambient temperature 25 degrees C.</Comment>
  </Header>
  <Luminaire>
    <Dimensions>
      <Length>1.20</Length>
      <Width>0.60</Width>
      <Height>0.10</Height>
    </Dimensions>
    <NumEmitter>1</NumEmitter>
  </Luminaire>
  <Equipment>
    </Gonioradiometer>
    <Type>IES _ C</Type>
    <MeasurementEquipment>LightLab</MeasurementEquipment>
    </Gonioradiometer>
    <IntegratingSphere>
      <MeasurementEquipment>LabSphere</MeasurementEquipment>
    </IntegratingSphere>
  </Equipment>
  <Emitter>
    <Quantity>1</Quantity>
    <Description>Integral LED module</Description>
    <CatalogNumber>Not applicable</CatalogNumber>
    <RatedLumens>0.0</RatedLumens>
    <InputWattage>35.4</InputWattage>
    <ColorTemperature>
      <FixedCCT>4000</FixedCCT>
    </ColorTemperature>
    <ColorRendering>
      <CIE _ CRI>

```

```

    <Ra>92</Ra>
    <R9>85</R9>
</CIE _ CRI>
<IES _ TM30>
    <Rf>60</Rf>
    <Rg>95</Rg>
</IES _ TM30>
</ColorRendering>
<LuminousData>
    <LuminousIntensity>
        <AbsolutePhotometry>true</AbsolutePhotometry>
        <Symm>Symm _ Full</Symm>
        <NumberMeasured>19</NumberMeasured>
        <NumberHorz>1</NumberHorz>
        <NumberVert>19</NumberVert>
        <IntData h="0.0" v="0.0">109</IntData>
        <IntData h="0.0" v="5.0">109</IntData>
        <IntData h="0.0" v="10.0">108</IntData>
        <IntData h="0.0" v="15.0">107</IntData>
        <IntData h="0.0" v="20.0">104</IntData>
        <IntData h="0.0" v="25.0">100</IntData>
        <IntData h="0.0" v="30.0">95</IntData>
        <IntData h="0.0" v="35.0">89</IntData>
        <IntData h="0.0" v="40.0">83</IntData>
        <IntData h="0.0" v="45.0">77</IntData>
        <IntData h="0.0" v="50.0">71</IntData>
        <IntData h="0.0" v="55.0">63</IntData>
        <IntData h="0.0" v="60.0">53</IntData>
        <IntData h="0.0" v="65.0">44</IntData>
        <IntData h="0.0" v="70.0">36</IntData>
        <IntData h="0.0" v="75.0">29</IntData>
        <IntData h="0.0" v="80.0">22</IntData>
        <IntData h="0.0" v="85.0">16</IntData>
        <IntData h="0.0" v="90.0">13</IntData>
    </LuminousIntensity>
    <LuminousFlux>1000.0</LuminousFlux>
</LuminousData>
<Channels>
    <NumChannels>3</NumChannels>
    <ChannelMult name="red">0.21</ChannelMult>
    <ChannelMult name="green">0.72</ChannelMult>
    <ChannelMult name="blue">0.08</ChannelMult>
</Channels>
<EmissionAreas>
    <BottomFace>
        <NumberBottom>2</NumberBottom>
        <BottomArea>

```

```

        <Length>0.50</Length>
        <Width>0.60</Width>
        <LengthOffset>-0.1</LengthOffset>
        <WidthOffset>0.0</WidthOffset>
    </BottomArea>
    <BottomArea>
        <Length>0.50</Length>
        <Width>0.60</Width>
        <LengthOffset>0.1</LengthOffset>
        <WidthOffset>0.0</WidthOffset>
    </BottomArea>
</BottomFace>
</EmissionAreas>
<EmitterCenter>
    <LengthOffset>0.00</LengthOffset>
    <WidthOffset>0.00</WidthOffset>
    <HeightOffset>-0.05</HeightOffset>
</EmitterCenter>
</Emitter>
</IESTM33-22>

```

A.2 Sample Luminous (Photometric) JSON Document

```

{
  "FileType": "IESTM33-22",
  "Version": 1.0,
  "Header": {
    "Manufacturer": "Academy Lighting",
    "CatalogNumber": "XET 55529",
    "Description": "LED 2' x 4' Troffer",
    "Laboratory": "Apex Analytics",
    "ReportNumber": "APEX-091101-004",
    "ReportDate": "2009-11-01",
    "DocumentCreator": "Apex Analytics",
    "UniqueIdentifier": "21EC2020-3AEA-4069-A2DD-08002B30309D",
    "Comment": [
      "Ambient temperature 25 degrees C."
    ]
  },
  "Luminaire": {
    "Dimensions": {
      "Length": 1.20,
      "Width": 0.60,
      "Height": 0.10
    },
    "NumEmitter": 1
  },
  "Equipment": {

```

```

"Gonioradiometer": {
  "Type": "IES_C",
  "MeasurementEquipment": [
    "LightLab"
  ]
},
"IntegratingSphere": {
  "MeasurementEquipment": [
    "LabSphere"
  ]
},
"Emitter": [
  {
    "Quantity": 1,
    "Description": "Integral LED module",
    "CatalogNumber": "Not applicable",
    "RatedLumens": 0.0,
    "InputWattage": 35.4,
    "ColorTemperature": {
      "FixedCCT": 4000
    },
    "ColorRendering": {
      "CIE_CRI": {
        "Ra": 92,
        "R9": 85
      },
      "IES_TM30": {
        "Rf": 60,
        "Rg": 95
      }
    },
    "LuminousData": {
      "LuminousIntensity": {
        "AbsolutePhotometry": true,
        "Symm": "Symm_Full",
        "NumberMeasured": 19,
        "NumberHorz": 1,
        "NumberVert": 19,
        "h": [ 0.0 ],
        "v": [ 0.0, 5.0, 10.0, 15.0, 20.0, 25.0, 30.0, 35.0,
          40.0, 45.0, 50.0, 55.0, 60.0, 65.0, 70.0, 75.0, 80.0,
          85.0, 90.0 ],
        "IntDataSymm": [ 109, 109, 108, 107, 104, 100, 95, 89, 83,
          77, 71, 63, 53, 44, 36, 29, 22, 16, 13 ]
      },
      "LuminousFlux": 1000.0
    }
  ]
}

```

```

    },
    "Channels": {
      "NumChannels": 3,
      "ChannelMult": [
        {
          "name": "red",
          "mult": 0.21
        },
        {
          "name": "green",
          "mult": 0.72
        },
        {
          "name": "blue",
          "mult": 0.08
        }
      ]
    },
    "EmissionAreas": {
      "BottomFace": {
        "NumberBottom": 2,
        "BottomArea": [
          {
            "Length": 0.50,
            "Width": 0.60,
            "LengthOffset": -0.1,
            "WidthOffset": 0.0
          },
          {
            "Length": 0.50,
            "Width": 0.60,
            "LengthOffset": 0.1,
            "WidthOffset": 0.0
          }
        ]
      }
    },
    "EmitterCenter": {
      "LengthOffset": 0.00,
      "WidthOffset": 0.00,
      "HeightOffset": -0.05
    }
  }
}

```

Annex B – Photon Flux

Note: Annex B is not part of ANSI/IES TM-33-23. It is included for information only.

Visible light is typically defined in terms of either lumens (luminous flux) or watts (radiant flux). Horticulturalists, however, measure electromagnetic radiation over the range of 280 nm to 800 nm in terms of photons per second.

Each photon is a quantum of electromagnetic energy, the magnitude of which is inversely proportional to its wavelength. Thus, the number of photons per second can be expressed in watts if the wavelength of each photon (or, more practically, of all the photons that have the same wavelength) is known.

Plant photosynthesis, however, depends on a chlorophyll molecule absorbing a single photon, regardless of its wavelength between approximately 400 nm and 700 nm. Thus, horticulturalists are not concerned with the photon's energy. They instead measure "photon flux" in terms of micromoles (6.022×10^{17}) of photons per second, $\mu\text{mol/s}$. If the wavelength range is 400 nm to 700 nm, this is photosynthetically active radiation (PAR).

Plants also respond to ultraviolet radiation (280 nm to 400 nm) and "far-red" radiation (700 nm to 800 nm). Depending on the application, horticultural measurements may be reported in terms of radiant flux or photon flux.

A comprehensive list of horticultural lighting metrics is provided in ANSI/ASABE S640 JUL2017, *Quantities and Units of Electromagnetic Radiation for Plants (Photosynthetic Organisms)* (see **Section 2.2**).

Annex C: Intensity Scaling & Multiplier

Note: Annex C is not part of ANSI/IES TM-33-23. It is included for information only.

The IntensityScaling boolean is informative and does not change or alter the interpretation of the values in the file. IntensityScaling can be used by the file writer to indicate that the candela values in the file were obtained by scaling test results in some manner.

A Multiplier instructs the reading software to multiply the values in the file by the Multiplier to obtain the 'real' values. The Multiplier is analogous to the Candela Multiplier in LM-63, and was used by some to 'scale' files for various wattages, lumen outputs, etc.

Because the IntensityScaling boolean is informative and tells what was done to the values in the file, and the Multiplier is instructive and modifies the values read from the file, there is no explicit or implicit connection between these two elements.

CandelaMultiplier wasn't always correctly implemented in IES file readers and there is no analogous value in Eulumdat files, so many manufacturers prefer to explicitly scale Intensity values and not rely upon the CandelaMultiplier.

Example 1: Light Bars

Take a luminaire comprised of some number of bars assembled into a frame. The manufacturer offers this in 8, 6 and 4 bar configurations. For each configuration the output per bar is identical.

**Option A: IntensityScaling - IntensityScaling = True, Multiplier = 1.0**

The manufacturer measures the output of the 8 bar fixture and creates a TM-33 for that product. They then take the IntData (candela values), and multiply them by 6/8 and 4/8 to create files for each configuration.

In this case, the IntensityScaling boolean can be used to identify that the derived files were scaled, and then the user can place a note in the Comment attribute to explain how this was done.

The Multiplier element will be set to 1.0 (or left out and defaulted to 1.0).

```
<IntensityScaling comment="8 bar fixture tested, scaled to 4 bar">true</IntensityScaling>
...
<Multiplier>1.0</Multiplier>
```

Option B: Multiplier - IntensityScaling = False, Multiplier not = 1.0

The manufacturer measures the output of the 8 bar fixture and creates a TM-33 for that product. To create the derived files, the manufacturer will use Multipliers of 6/8 and 4/8, leaving the originally measured IntData elements as they are.

In this case the IntensityScaling boolean would remain False, since the IntData elements in the file have not been scaled in any way before being placed in the file.

```
<IntensityScaling >false</IntensityScaling>
...
<Multiplier comment="8 bar fixture tested, scaled to 4 bar">0.5</Multiplier>
```

Example 2: CRI & CCT variations

Take a luminaire with various CCT and CRI options, say 3000K and 5000K, 80 and 90 CRI options. The manufacturer has obtained integrating sphere data to show that the following lumen outputs occur for the entire product:

	80 CRI	90 CRI
3000K	12,000 lumens	11,000 lumens
5000K	14,000 lumens	13,000 lumens

This allows the manufacturer to determine the following scale relationship between the lumen output, setting the 3000K, 90CRI as the reference, typically the worst case for efficacy.

	80 CRI	90 CRI
3000K	109.09%	100.00%
5000K	127.27%	118.18%

The manufacturer may then perform an LM-79 goniometric test on the 3000K, 90 CRI version and then wants to generate TM-33 files for the remaining 3 options.

Option A: IntensityScaling - IntensityScaling = True, Multiplier = 1.0

The IntData (candela values) elements will each be multiplied by the appropriate scale factor and written into the new TM-33 files.

The IntensityScaling boolean can be used to identify that the derived files were scaled, and then the user can place a note in the Comment element to explain how this was done.

The Multiplier element will be set to 1.0 (or left out and defaulted to 1.0).

Option B: Multiplier - IntensityScaling = False, Multiplier not = 1.0

To create the derived files, the manufacturer will use Multipliers of 109.09%, 127.27%, and 118.18%, leaving the originally measured IntData elements as they are.

In this case the IntensityScaling boolean would remain False, since the IntData elements in the file have not been scaled in any way before being placed in the file.

Example 3: Measurement Units - IntensityScaling = True, Multiplier = Not 1.0

Take a very low output product. The lab equipment is very precise and records the output in Cd, but the resulting values are actually on the order to micro-Cd, so 0.000001.

To reduce file size, the lab may choose to convert all of the Cd to micro-Cd and write micro-Cd into the TM-33 file by scaling everything by 1,000,000. They would indicate that this scaling had been done by setting IntensityScaling to True and include a Comment.

Because the reading software is expecting Cd, not micro-Cd, the Multiplier element would be set to 0.000001.

Example 4: Eulumdat file as source data

Eulumdat files contain Intensity values as cd/klm.

Option A: Retain reported values - IntensityScaling = True, Multiplier = Not 1.0

When converting an LDT file to TM-33, reading in Intensity values as cd/klm, one option is to write out the cd/klm values into the IntData elements. Because TM-33 expects cd, not cd/klm, the Multiplier element can be set to OutputLumens / 1000. This then instructs the reading software to convert cd/klm to cd for further use.

The IntensityScaling here should be set to True since the IntData values in the file are scaled to cd/klm, not the actually measured cd.

Option B: Scale reported values - IntensityScaling = False, Multiplier = 1.0

When converting an LDT file to TM-33, reading in Intensity values as cd/klm, the second option is to write out the cd values into the IntData elements by scaling the cd/klm values.

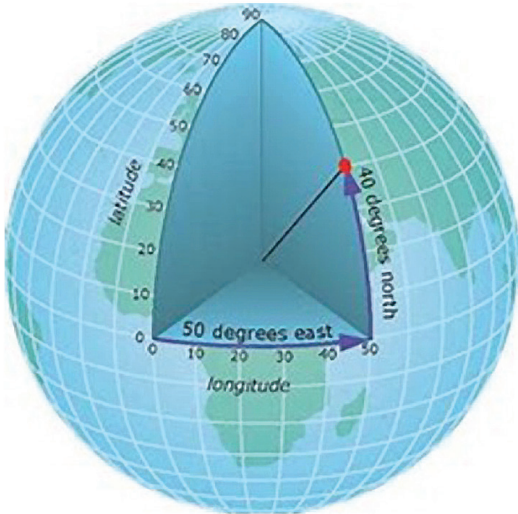
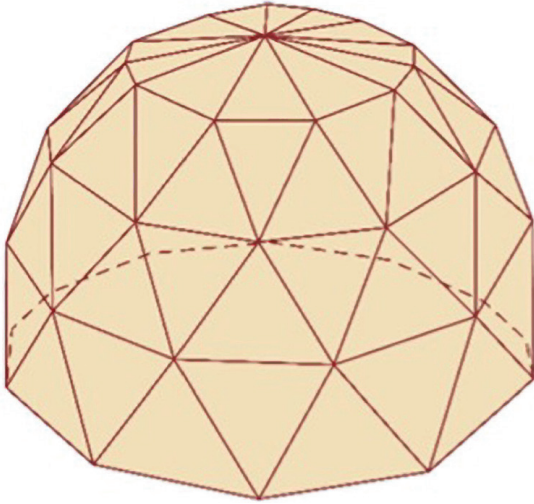
Since cd values were originally measured and the TM-33 contains candela values, the IntensityScaling should be set to False and the Multiplier set to 1.0.

Annex D: Symmetry for Type C Coordinate Systems

Note: Annex D is not part of ANSI/IES TM-33-23. It is included for information only.

In a traditional Type C photometric web, for every horizontal angle, there is a matching number of vertical data points (For H=0 if there are 37 vertical positions, then for H=90 there also will be 37 vertical positions that match the same vertical angles as H=0), since the system is a regular polar coordinate system.

For Arbitrary symmetric, there is no regularity of the H/V sets, and so you may have H=0 with 37 vertical values, and H=90 with 12 vertical values, or they may both have 37 vertical values, but at varying angle positions. An example of this would be the geodesic dome shown below.

Regular Polar Coordinates	Irregular Coordinates
	
In the traditional polar coordinate systems, at a Horizontal angle of 0 and a horizontal angle of 30, the same number and location of vertical angles will be reported. H=0, V=0,10,20,30,40,50,60,70,80,90 H=30, V=0,10,20,30,40,50,60,70,80,90	In a Symm_Abitrary coordinate system, the location of the data points are not in a traditional polar coordinate grid. A set of points may be as follows: H=0, V=0, 45, 90 H=10, V=22.5, 67.5, 90

Plotting Symm_Arbitrary data is still possible since the data still contain polar coordinate locations, however, that data is not in regular positions, and so each plane may contain a different number of data points or not contain the same start and end points. The plotting would be just like traditional Type C plotting, where you take a given horizontal angle (H=0) and then at each vertical angle that is in the data set, plot the intensity and connect them via a line. Then proceed to do this for every H angle you find in the data. If the data is sparse, this could lead to odd plots, but that is already the case with sparse data in traditional angle sets.

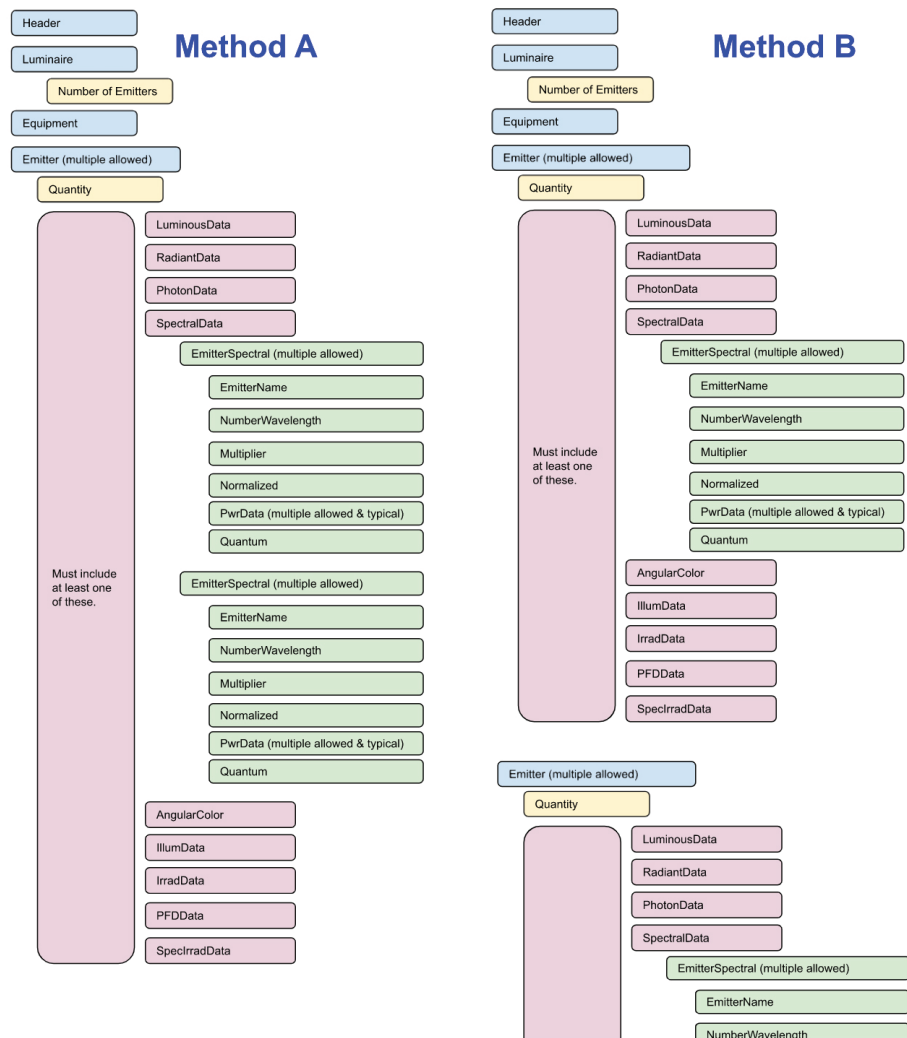
Annex E: Emitter Quantities

Note: Annex E is not part of ANSI/IES TM-33-23. It is included for information only.

TM-33 contains two potential approaches for including multiple sets of data within a single file. This would commonly be done for products which have two or more unique output levels, distributions, or spectrums of light, such as circadian lighting.

Method A: Multiple Spectral Data Sets are contained within one Emitter element.

Method B: Each Emitter element contains one set of spectral data and there are multiple Emitter elements.



Method A allows multiple spectra, but if other elements are included (Luminous Data, Radiant Data, IllumData), these are only included once, implying that they are the same regardless of spectra.

Method B allows for spectra to be matched with other items like Luminous Data under an Emitter element. This allows the distribution of light to change with the spectra by including more than one Emitter element. Method B is the preferred method.

	Approach A	Approach B
Description	Multiple EmitterSpectral's for a single Emitter	Multiple Emitters, each with a single EmitterSpectral
Example	Luminaire with RGBW LED chips, so there is 1 Emitter type, which has multiple channels within it, so these are each contained within an EmitterSpectral element. For the Quantity, the # of LEDs could be used, or the # of channels, there is no requirement here.	Luminaire with a cool white and a warm white LED array. Since these are separate arrays, including them each as their own Emitter with their own EmitterSpectral data may make the most sense.
NumEmitter	1	2
Quantity	56 LEDs, or 4 Channels as examples	56 LEDs for WW and 50 for CW as an example, or 1 and 1 (most common coming from absolute IES files for LED fixtures).

Process for Change to an ANSI/IES Standard under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

- a) Proposed change accepted for public review without modification
- b) Proposed change accepted for public review with modification
- c) Proposed change accepted for further study
- d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

NOTE: Use a separate form for each comment. Submit to the Director of Standards, IES, 120 Wall Street, 17th Floor, New York, NY 10005-4001. Email: standards@ies.org. Fax: 212-248-5017.

1. Submitter: _____
Affiliation: _____
Address: _____
City: _____ State: _____ Zip: _____ Country: _____
Telephone: _____
Fax: _____
E-mail: _____

I hereby grant the Illuminating Engineering Society (IES) the non-exclusive royalty rights, including non-exclusive rights in copyright, in my proposals. I understand that I acquire no rights in publication of the standard in which my proposals in this, or other analogous, form are used. I hereby attest that I have the authority and am empowered to grant this copyright release.

Submitter's signature: _____ Date: _____

2. Title of publications and year published _____

3. Clause (section), sub-clause or paragraph number; and page number: _____

4. My proposal (check one):

- ☐ Change to read as follows
- ☐ Delete and substitute as follows
- ☐ Add new text as follows
- ☐ Delete without substitution

Use underscore to show material to be added (added) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

6. Reason and substantiation:

Select as applicable:

- ☐ Additional pages are attached. Number of additional pages: _____
- ☐ Attachments or referenced materials cited in this proposal accompany this proposed change.

Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:



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